

SSD0323

Advance Information

**128 x 80, 16 Gray Scale Dot Matrix
OLED/PLED Segment/Common Driver with Controller**

This document contains information on a new product. Specifications and information herein are subject to change without notice.

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GENERAL DESCRIPTION

SSD0323 is a single-chip CMOS OLED/PLED driver with controller for organic/polymer light emitting diode dot-matrix graphic display system. SSD0323 consists of 208 high voltage/current driving output pins for driving 128 segments and 80 commons. This IC is designed for Common Cathode type OLED/PLED panel.

SSD0323 displays data directly from its internal 128x80x4 bits Graphic Display Data RAM (GDDRAM). Data/Commands are sent from general MCU through the hardware selectable 6800-/8080-series compatible Parallel Interface or Serial Peripheral Interface.

SSD0323 has a 128-step contrast control and a 16 gray level control. The embedded on-chip oscillator and DC-DC voltage converter reduce the number of external components.

FEATURES

- Support max. 128 x 80 matrix panel
- Power supply: VDD=2.4V - 3.5V
VCC=8.0V - 16.0V
- OLED driving output voltage, 14V maximum
- DC-DC voltage converter
- Segment maximum source current: 300uA
- Common maximum sink current: 40mA
- Embedded 128 x 80 x 4 bit SRAM display memory
- External current reference
- 128 step contrast control on monochrome passive OLED panel
- 16 gray scale
- On-Chip Oscillator
- Programmable Frame Rate/Pre-charge voltage and segment low voltage (VSL)
- 8-bit 6800-series Parallel Interface, 8-bit 8080-series Parallel Interface, Serial Peripheral Interface.
- Row re-mapping and Column re-mapping
- Low power consumption (<5.0uA @sleep mode)
- Wide range of operating temperature: -40 to 85 °C

ORDERING INFORMATION

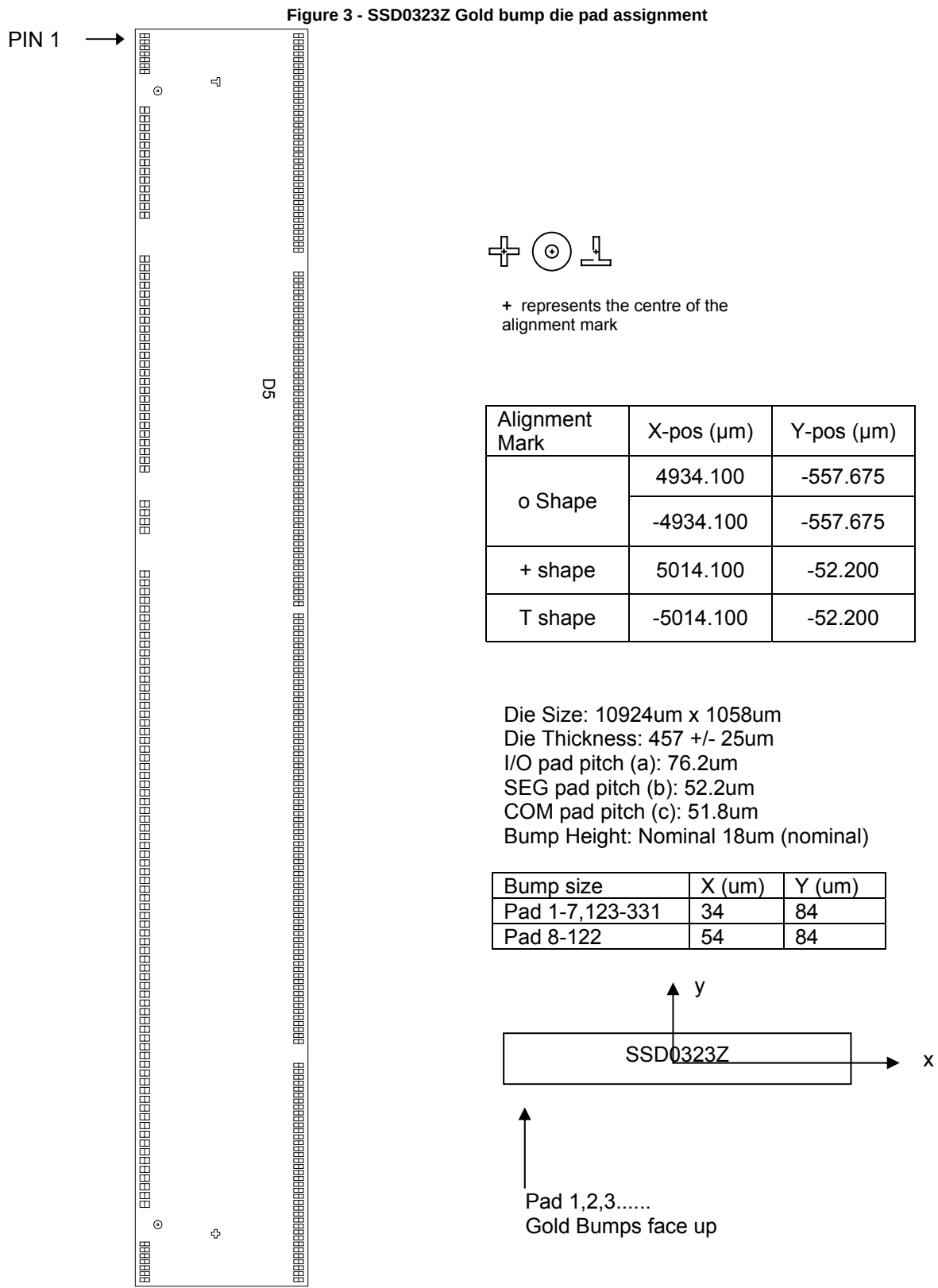
Table 1 - Ordering Information

Ordering Part Number	SEG	COM	Package Form	Reference
SSD0323Z	128	80	COG	Page 8

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SSD0323Z GOLD BUMP DIE PAD ASSIGNMENT



SSD0323Z Die Pad Coordinates

Pad#	Signal	X-pos	Y-pos	Pad#	Signal	X-pos	Y-pos	Pad#	Signal	X-pos	Y-pos	Pad#	Signal	X-pos	Y-pos
1	DUMMY	-5414.000	-672.075	91	VDD	2395.725	-672.075	181	SEG14	2610.000	672.075	271	SEG104	-2140.200	672.075
2	DUMMY	-5361.800	-672.075	92	ICAS	2471.925	-672.075	182	SEG15	2557.800	672.075	272	SEG105	-2192.400	672.075
3	COM75	-5309.600	-672.075	93	IREF	2548.125	-672.075	183	SEG16	2505.600	672.075	273	SEG106	-2244.600	672.075
4	COM76	-5257.800	-672.075	94	VCOMH	2624.325	-672.075	184	SEG17	2453.400	672.075	274	SEG107	-2296.800	672.075
5	COM77	-5206.000	-672.075	95	VCOMH	2700.525	-672.075	185	SEG18	2401.200	672.075	275	SEG108	-2349.000	672.075
6	COM78	-5154.200	-672.075	96	VREF	2776.725	-672.075	186	SEG19	2349.000	672.075	276	SEG109	-2401.200	672.075
7	COM79	-5102.400	-672.075	97	VCC	2852.925	-672.075	187	SEG20	2296.800	672.075	277	SEG110	-2453.400	672.075
8	DUMMY	-4767.075	-672.075	98	VCC	2929.125	-672.075	188	SEG21	2244.600	672.075	278	SEG111	-2505.600	672.075
9	DUMMY	-4690.875	-672.075	99	VDD	3005.325	-672.075	189	SEG22	2192.400	672.075	279	SEG112	-2557.800	672.075
10	DUMMY	-4614.675	-672.075	100	VSL	3081.525	-672.075	190	SEG23	2140.200	672.075	280	SEG113	-2610.000	672.075
11	DUMMY	-4538.475	-672.075	101	VSS	3157.725	-672.075	191	SEG24	2088.000	672.075	281	SEG114	-2662.200	672.075
12	DUMMY	-4462.275	-672.075	102	VCL	3233.925	-672.075	192	SEG25	2035.800	672.075	282	SEG115	-2714.400	672.075
13	DUMMY	-4386.075	-672.075	103	VCL	3310.125	-672.075	193	SEG26	1983.600	672.075	283	SEG116	-2766.600	672.075
14	DUMMY	-4309.875	-672.075	104	VCL	3386.325	-672.075	194	SEG27	1931.400	672.075	284	SEG117	-2818.800	672.075
15	DUMMY	-4233.675	-672.075	105	DUMMY	3462.525	-672.075	195	SEG28	1879.200	672.075	285	SEG118	-2871.000	672.075
16	DUMMY	-4157.475	-672.075	106	DUMMY	3538.725	-672.075	196	SEG29	1827.000	672.075	286	SEG119	-2923.200	672.075
17	DUMMY	-4081.275	-672.075	107	DUMMY	3614.925	-672.075	197	SEG30	1774.800	672.075	287	SEG120	-2975.400	672.075
18	DUMMY	-4005.075	-672.075	108	DUMMY	3691.125	-672.075	198	SEG31	1722.600	672.075	288	SEG121	-3027.600	672.075
19	DUMMY	-3928.875	-672.075	109	DUMMY	3767.325	-672.075	199	SEG32	1670.400	672.075	289	SEG122	-3079.800	672.075
20	DUMMY	-3852.675	-672.075	110	DUMMY	3843.525	-672.075	200	SEG33	1618.200	672.075	290	SEG123	-3132.000	672.075
21	VCL	-3471.675	-672.075	111	DUMMY	3919.725	-672.075	201	SEG34	1566.000	672.075	291	SEG124	-3184.200	672.075
22	VCL	-3395.475	-672.075	112	DUMMY	3995.925	-672.075	202	SEG35	1513.800	672.075	292	SEG125	-3236.400	672.075
23	VCL	-3319.275	-672.075	113	DUMMY	4072.125	-672.075	203	SEG36	1461.600	672.075	293	SEG126	-3288.600	672.075
24	VSS	-3243.075	-672.075	114	DUMMY	4148.325	-672.075	204	SEG37	1409.400	672.075	294	SEG127	-3340.800	672.075
25	VSSB	-3166.875	-672.075	115	DUMMY	4224.525	-672.075	205	SEG38	1357.200	672.075	295	COM40	-3548.400	672.075
26	VSSB	-3090.675	-672.075	116	DUMMY	4300.725	-672.075	206	SEG39	1305.000	672.075	296	COM41	-3600.200	672.075
27	VSL	-3014.475	-672.075	117	DUMMY	4376.925	-672.075	207	SEG40	1252.800	672.075	297	COM42	-3652.000	672.075
28	VSLCAP	-2938.275	-672.075	118	DUMMY	4453.125	-672.075	208	SEG41	1200.600	672.075	298	COM43	-3703.800	672.075
29	VSLCAP	-2862.075	-672.075	119	DUMMY	4529.325	-672.075	209	SEG42	1148.400	672.075	299	COM44	-3755.600	672.075
30	VDD	-2785.875	-672.075	120	DUMMY	4605.525	-672.075	210	SEG43	1096.200	672.075	300	COM45	-3807.400	672.075
31	VCC	-2709.675	-672.075	121	DUMMY	4681.725	-672.075	211	SEG44	1044.000	672.075	301	COM46	-3859.200	672.075
32	VCC	-2633.475	-672.075	122	DUMMY	4757.925	-672.075	212	SEG45	991.800	672.075	302	COM47	-3911.000	672.075
33	VCOMH	-2557.275	-672.075	123	COM39	5102.400	-672.075	213	SEG46	939.600	672.075	303	COM48	-3962.800	672.075
34	VCOMH	-2481.075	-672.075	124	COM38	5154.200	-672.075	214	SEG47	887.400	672.075	304	COM49	-4014.600	672.075
35	IR8	-2404.875	-672.075	125	COM37	5206.000	-672.075	215	SEG48	835.200	672.075	305	COM50	-4066.400	672.075
36	IR7	-2328.675	-672.075	126	COM36	5257.800	-672.075	216	SEG49	783.000	672.075	306	COM51	-4118.200	672.075
37	IR6	-2252.475	-672.075	127	COM35	5309.600	-672.075	217	SEG50	730.800	672.075	307	COM52	-4170.000	672.075
38	IR5	-2176.275	-672.075	128	DUMMY	5361.800	-672.075	218	SEG51	678.600	672.075	308	COM53	-4221.800	672.075
39	IR4	-2100.075	-672.075	129	DUMMY	5414.000	-672.075	219	SEG52	626.400	672.075	309	COM54	-4273.600	672.075
40	IR3	-2023.875	-672.075	130	DUMMY	5414.000	672.075	220	SEG53	574.200	672.075	310	COM55	-4325.400	672.075
41	IR2	-1947.675	-672.075	131	DUMMY	5361.800	672.075	221	SEG54	522.000	672.075	311	COM56	-4377.200	672.075
42	IR1	-1871.475	-672.075	132	COM34	5309.600	672.075	222	SEG55	469.800	672.075	312	COM57	-4429.000	672.075
43	IR0	-1795.275	-672.075	133	COM33	5257.800	672.075	223	SEG56	417.600	672.075	313	COM58	-4480.800	672.075
44	VSS	-1719.075	-672.075	134	COM32	5206.000	672.075	224	SEG57	365.400	672.075	314	COM59	-4532.600	672.075
45	VSSB	-1642.875	-672.075	135	COM31	5154.200	672.075	225	SEG58	313.200	672.075	315	COM60	-4584.400	672.075
46	GDR	-1338.075	-672.075	136	COM30	5102.400	672.075	226	SEG59	261.000	672.075	316	COM61	-4636.200	672.075
47	GDR	-1261.875	-672.075	137	COM29	5050.600	672.075	227	SEG60	208.800	672.075	317	COM62	-4688.000	672.075
48	GDR	-1185.675	-672.075	138	COM28	4998.800	672.075	228	SEG61	156.600	672.075	318	COM63	-4739.800	672.075
49	GDR	-1109.475	-672.075	139	COM27	4947.000	672.075	229	SEG62	104.400	672.075	319	COM64	-4791.600	672.075
50	VDOB	-728.475	-672.075	140	COM26	4895.200	672.075	230	SEG63	52.200	672.075	320	COM65	-4843.400	672.075
51	VDOB	-652.275	-672.075	141	COM25	4843.400	672.075	231	SEG64	0.000	672.075	321	COM66	-4895.200	672.075
52	VDD	-576.075	-672.075	142	COM24	4791.600	672.075	232	SEG65	-52.200	672.075	322	COM67	-4947.000	672.075
53	VDD	-499.875	-672.075	143	COM23	4739.800	672.075	233	SEG66	-104.400	672.075	323	COM68	-4998.800	672.075
54	FB	-423.675	-672.075	144	COM22	4688.000	672.075	234	SEG67	-156.600	672.075	324	COM69	-5050.600	672.075
55	RESE	-347.475	-672.075	145	COM21	4636.200	672.075	235	SEG68	-208.800	672.075	325	COM70	-5102.400	672.075
56	VREF	-271.275	-672.075	146	COM20	4584.400	672.075	236	SEG69	-261.000	672.075	326	COM71	-5154.200	672.075
57	BGGND	-195.075	-672.075	147	COM19	4532.600	672.075	237	SEG70	-313.200	672.075	327	COM72	-5206.000	672.075
58	VSS	-118.875	-672.075	148	COM18	4480.800	672.075	238	SEG71	-365.400	672.075	328	COM73	-5257.800	672.075
59	VCC	-42.675	-672.075	149	COM17	4429.000	672.075	239	SEG72	-417.600	672.075	329	COM74	-5309.600	672.075
60	GPI0	33.525	-672.075	150	COM16	4377.200	672.075	240	SEG73	-469.800	672.075	330	DUMMY	-5361.800	672.075
61	GPI01	109.725	-672.075	151	COM15	4325.400	672.075	241	SEG74	-522.000	672.075	331	DUMMY	-5414.000	672.075
62	VDD	185.925	-672.075	152	COM14	4273.600	672.075	242	SEG75	-574.200	672.075				
63	BS0	262.125	-672.075	153	COM13	4221.800	672.075	243	SEG76	-626.400	672.075				
64	VSS	338.325	-672.075	154	COM12	4170.000	672.075	244	SEG77	-678.600	672.075				
65	BS1	414.525	-672.075	155	COM11	4118.200	672.075	245	SEG78	-730.800	672.075				
66	VDD	490.725	-672.075	156	COM10	4066.400	672.075	246	SEG79	-783.000	672.075				
67	BS2	566.925	-672.075	157	COM9	4014.600	672.075	247	SEG80	-835.200	672.075				
68	VSS	643.125	-672.075	158	COM8	3962.800	672.075	248	SEG81	-887.400	672.075				
69	FR	719.325	-672.075	159	COM7	3911.000	672.075	249	SEG82	-939.600	672.075				
70	CL	795.525	-672.075	160	COM6	3859.200	672.075	250	SEG83	-991.800	672.075				
71	DOF#	871.725	-672.075	161	COM5	3807.400	672.075	251	SEG84	-1044.000	672.075				
72	VSS	947.925	-672.075	162	COM4	3755.600	672.075	252	SEG85	-1096.200	672.075				
73	CS#	1024.125	-672.075	163	COM3	3703.800	672.075	253	SEG86	-1148.400	672.075				
74	RES#	1100.325	-672.075	164	COM2	3652.000	672.075	254	SEG87	-1200.600	672.075				
75	DY#	1176.525	-672.075	165	COM1	3600.200	672.075	255	SEG88	-1252.800	672.075				
76	VSS	1252.725	-672.075	166	COM0	3548.400	672.075	256	SEG89	-1305.000	672.075				
77	R/W#(WR#)	1328.925	-672.075	167	SEG0										

PIN DESCRIPTION

M, DOF#

These pins are No Connection pins. Nothing should be connected to these pins, nor they are connected together. These pins should be left open individually.

CL

This pin is the system clock input. When internal clock is enabled (i.e. CLS pin is pulled HIGH), this pin should be left open. Nothing should be connected to this pin. In this case, the output clock frequency equals to the internal clock frequency. When internal oscillator is disabled (i.e. CLS pin is pulled HIGH), this pin receives display clock signal from external clock source.

M/S#

This pin is an input pin and must be pulled HIGH to enable the chip function.

CLS

This pin is internal clock enable. When this pin is pulled HIGH, internal clock is enabled. The internal clock will be disabled when it is pulled LOW, an external clock source must be connected to CL pin for normal operation.

BS0, BS1, BS2

These pins are MCU interface selection input. See the following table:

Table 3 - MCU interface setting

	6800-parallel interface	8080-parallel interface	Serial interface
BS0	0	0	0
BS1	0	1	0
BS2	1	1	0

CS#

This pin is the chip select input. The chip is enabled for MCU communication only when CS# is pulled LOW.

RES#

This pin is reset signal input. When the pin is LOW, initialization of the chip is executed.

D/C#

This pin is Data/Command control pin. When the pin is pulled HIGH, the input at D₇-D₀ is treated as display data. When the pin is pulled LOW, the input at D₇-D₀ will be transferred to the command register. For detail relationship to MCU interface signals, please refer to the Timing Characteristics Diagrams.

R/W# (WR#)

This pin is MCU interface input. When interfacing to a 6800-series microprocessor, this pin will be used as Read/Write (R/W#) selection input. Pull this pin to "HIGH" for read mode and pull it to "LOW" for write mode. When 8080 interface mode is selected, this pin will be the Write (WR#) input. Data write operation is initiated when this pin is pulled LOW and the CS# is pulled LOW.

E (RD#)

This pin is MCU interface input. When interfacing to a 6800-series microprocessor, this pin will be used as the Enable (E) signal. Read/write operation is initiated when this pin is pulled HIGH and the CS# is pulled LOW. When connecting to an 8080-microprocessor, this pin receives the Read (RD#) signal. Data read operation is initiated when this pin is pulled LOW and CS# is pulled LOW.

D₇-D₀

These pins are 8-bit bi-directional data bus to be connected to the microprocessor's data bus. When serial mode is selected, D₁ will be the serial data input SDIN and D₀ will be the serial clock input SCLK.

V_{DD}

This is a voltage supply pin. It must be connected to external source.

V_{SS}

This is a ground pin. It also acts as a reference for the logic pins and the OLED driving voltages. It must be connected to external ground.

V_{CC}

This is the most positive voltage supply pin of the chip. It can be supplied externally or generated internally by using internal DC-DC voltage converter.

V_{REF}

This pin is the voltage reference for pre-charge voltage in driving OLED device. Voltage should be set to match with the OLED driving voltage in current drive phase. It can be either supplied externally or connected to V_{CC}.

I_{REF}

This pin is segment current reference pin. A resistor should be connected between this pin and V_{SS}. Set the current at 10uA.

V_{COMH}

A capacitor should be connected between this pin and V_{SS}

VSL

This pin is the output pin for the voltage output low level for SEG signals. A capacitor should be connected between this pin and V_{SS}.

VSLCAP

This pin is the output pin for the voltage output low level for SEG signals. A capacitor should be connected between this pin and V_{SS}.

VCL

This pin is the output pin for the voltage output low level for COM signals. This pin should be connected to V_{SS}.

VDDB

This is the power supply pin for the GDR pin buffer. It must be connected when the converter is used.

VSSB

This is the GND pin for the GDR pin buffer. It must be connected when the converter is used.

GDR

This output pin drives the gate of the external NMOS of the booster circuit.

FB

This pin is the feedback resistor input of the booster circuit. It is used to adjust the booster output voltage level(Vcc).

RESE

This pin connects to the source current pin of the external NMOS of the booster circuit.

VB_{REF}

This pin is the internal voltage reference of booster circuit. A stabilization capacitor should be connected between this pin and Vss for both internal and external VCC usage.

COM0-COM79

These pins provide the Common switch signals to the OLED panel. These pins are in high impedance state when display is OFF.

SEG0-SEG127

These pins provide the OLED segment driving signals. These pins are in high impedance state when display is OFF.

NC

These pins should be left open individually.

FUNCTIONAL BLOCK DESCRIPTIONS

Command Decoder

This module determines whether the input data is interpreted as data or command. Data is interpreted based upon the input of the D/C# pin.

If D/C# pin is HIGH, the input at D₇-D₀ is written to Graphic Display Data RAM (GDDRAM). If it is LOW, the input at D₇-D₀ is interpreted as a Command which will be decoded and be written to the corresponding command register.

MPU Interface selection

SSD0323 MCU interface consist of 8 data pins and 5 control pins. The pin assignment at different interface mode is summarized in Table below. Different MCU mode can be set by hardware selection on BS[2:0] pins (please refer to Table 3 for BS[2:0] setting).

Table 4 : MCU interface assignment under different bus interface mode

Pin Name Bus Interface	Data/Command Interface								Control Signal				
	D7	D6	D5	D4	D3	D2	D1	D0	E	R/W#	CS#	D/C#	RES#
8-bit 8080	D[7:0]								RD#	WR#	CS#	D/C#	RES#
8-bit 6800	D[7:0]								E	R/W#	CS#	D/C#	RES#
SPI	Tie LOW					NC	SDIN	SCLK	Tie LOW		CS#	D/C#	RES#

MPU Parallel 6800-series Interface

The parallel interface consists of 8 bi-directional data pins (D[7:0]), R/W#, D/C#, E and CS#.

A LOW in R/W# indicates WRITE operation and HIGH in R/W# indicates READ operation.

A LOW in D/C# indicates COMMAND read/write and HIGH in D/C# indicates DATA read/write.

The E input serves as data latch signal while CS# is LOW. Data is latched at the falling edge of E signal.

Table 5: Control pins of 6800 interface

Function	E	R/W#	CS#	D/C#
Write command	↓	L	L	L
Read status	↓	H	L	L
Write data	↓	L	L	H
Read data	↓	H	L	H

Note

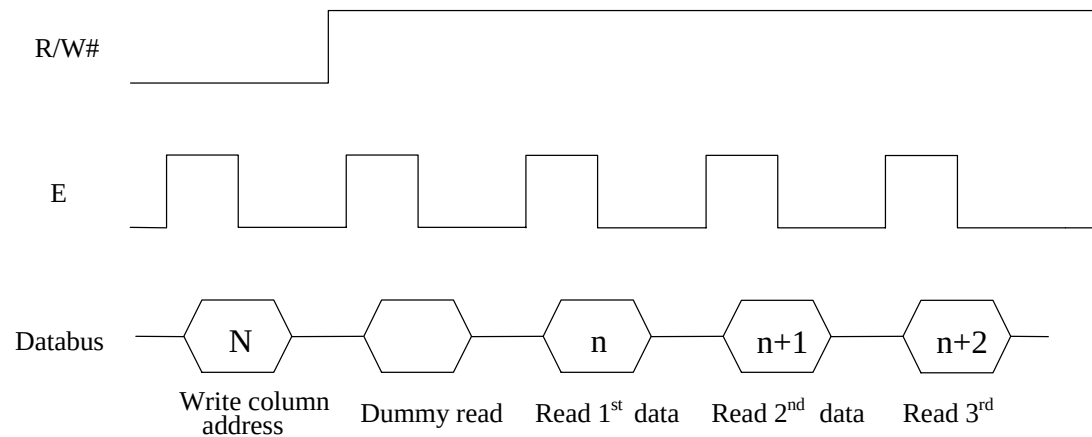
⁽¹⁾↓ stands for falling edge of signal

H stands for HIGH in signal

L stands for LOW in signal

In order to match the operating frequency of display RAM with that of the microprocessor, some pipeline processing is internally performed which requires the insertion of a dummy read before the first actual display data read. This is shown in Figure 4.

Figure 4 : Data read back procedure - insertion of dummy read



MPU Parallel 8080-series Interface

The parallel interface consists of 8 bi-directional data pins (D[7:0]), RD#, WR#, D/C# and CS#.

A LOW in D/C# indicates COMMAND read/write and HIGH in D/C# indicates DATA read/write.

A rising edge of RD# input serves as a data READ latch signal while CS# is kept LOW.

A rising edge of WR# input serves as a data/command WRITE latch signal while CS# is kept LOW.

Figure 5 : Example of Write procedure in 8080 parallel interface mode

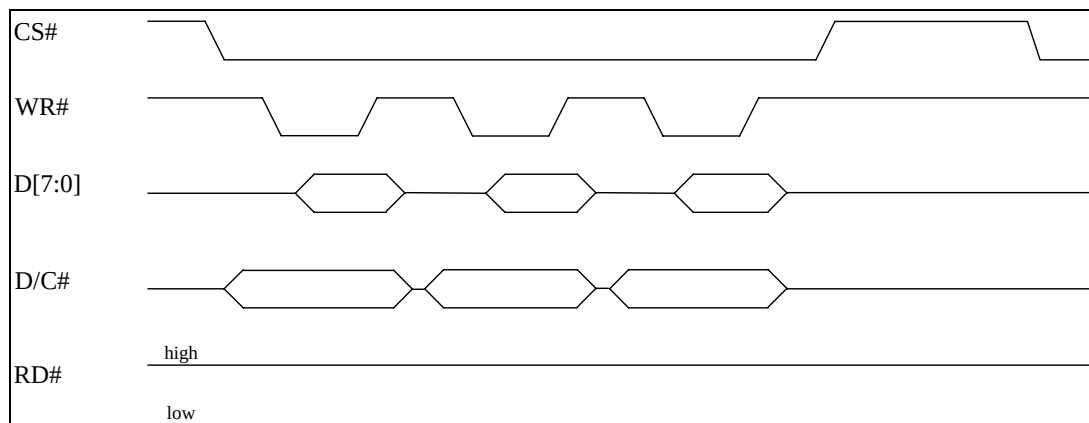


Figure 6 : Example of Read procedure in 8080 parallel interface mode

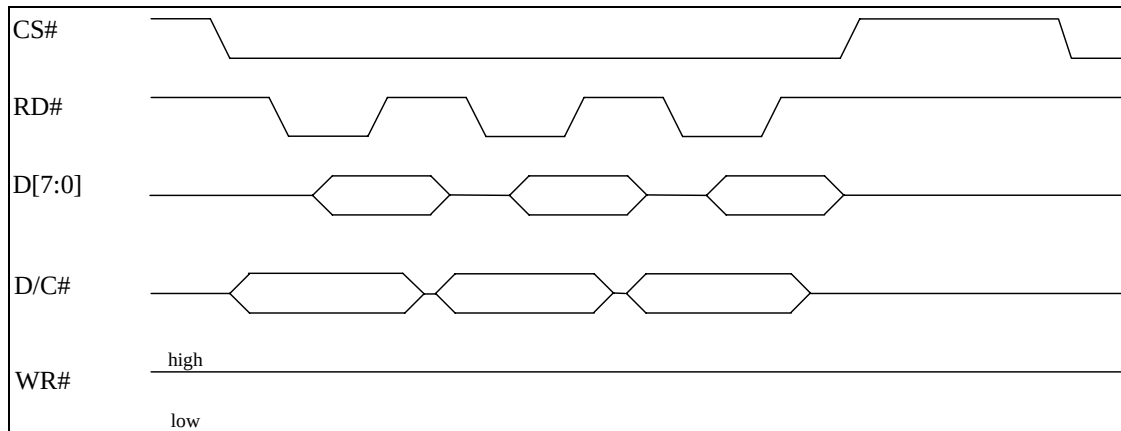


Table 6: Control pins of 8080 interface (Form 1)

Function	RD#	WR#	CS#	D/C#
Write command	H	↑	L	L
Read status	↑	H	L	L
Write data	H	↑	L	H
Read data	↑	H	L	H

Note

⁽¹⁾ ↑ stands for rising edge of signal

⁽²⁾ H stands for HIGH in signal

⁽³⁾ L stands for LOW in signal

⁽⁴⁾ Refer to Figure 21 for Form 1 8080-Series MPU Parallel Interface Timing Characteristics

Alternatively, RD# and WR# can be keep stable while CS# serves as the data/command latch signal.

Table 7: Control pins of 8080 interface (Form 2)

Function	RD#	WR#	CS#	D/C#
Write command	H	L	↑	L
Read status	L	H	↑	L
Write data	H	L	↑	H
Read data	L	H	↑	H

Note

⁽¹⁾ ↑ stands for rising edge of signal

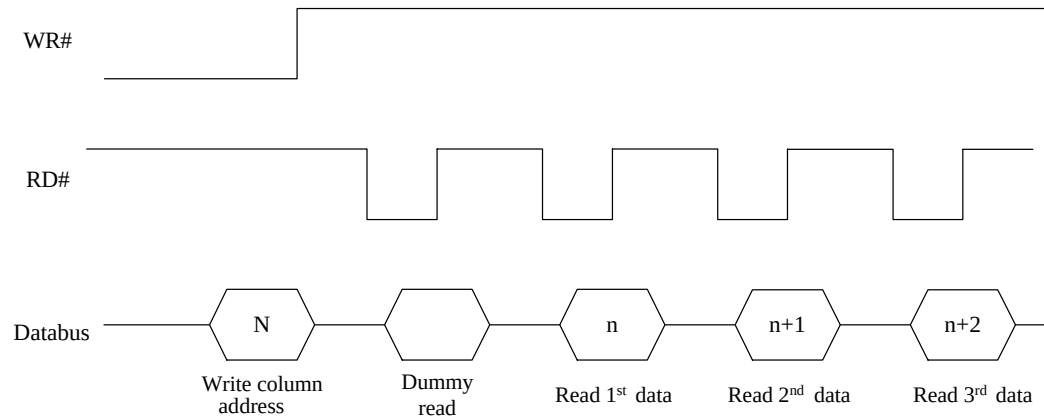
⁽²⁾ H stands for HIGH in signal

⁽³⁾ L stands for LOW in signal

⁽⁴⁾ Refer to Figure 22 for Form 2 8080-Series MPU Parallel Interface Timing Characteristics

In order to match the operating frequency of display RAM with that of the microprocessor, some pipeline processing is internally performed which requires the insertion of a dummy read before the first actual display data read. This is shown in Figure 7.

Figure 7: Display data read back procedure - insertion of dummy read



MPU Serial Interface

The serial interface consists of serial clock SCLK, serial data SDIN, D/C#, CS#. In SPI mode, D0 acts as SCLK, D1 acts as SDIN. For the unused data pins, D2 should be left open. The pins from D3 to D7, E and R/W# can be connected to an external ground.

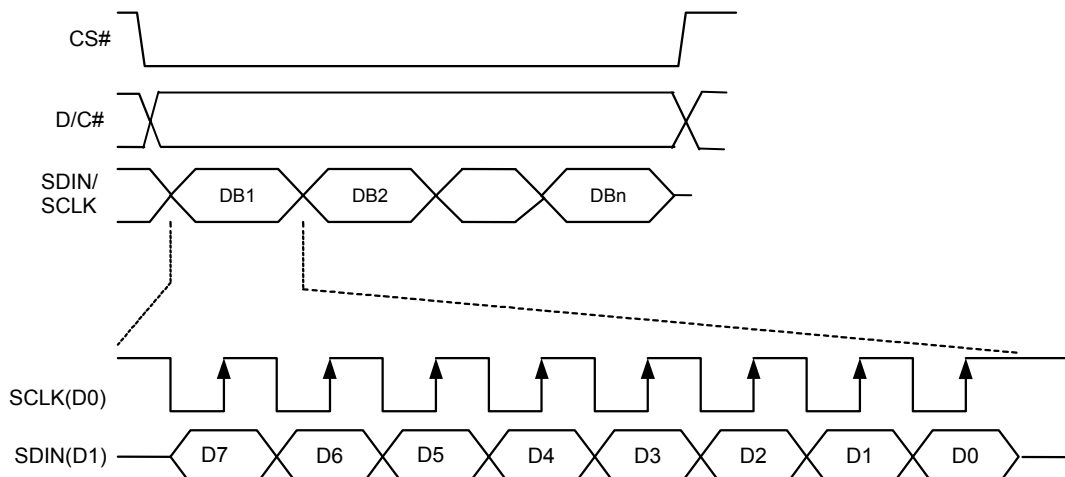
Table 8: Control pins of Serial interface

Function	E	R/W#	CS#	D/C#
Write command	Tie LOW	Tie LOW	L	L
Write data	Tie LOW	Tie LOW	L	H

SDIN is shifted into an 8-bit shift register on every rising edge of SCLK in the order of D7, D6, ... D0. D/C# is sampled on every eighth clock and the data byte in the shift register is written to the Graphic Display Data RAM (GDDRAM) or command register in the same clock.

Under serial mode, only write operations are allowed.

Figure 8: Display data write procedure in SPI mode



Oscillator Circuit and Display Time Generator

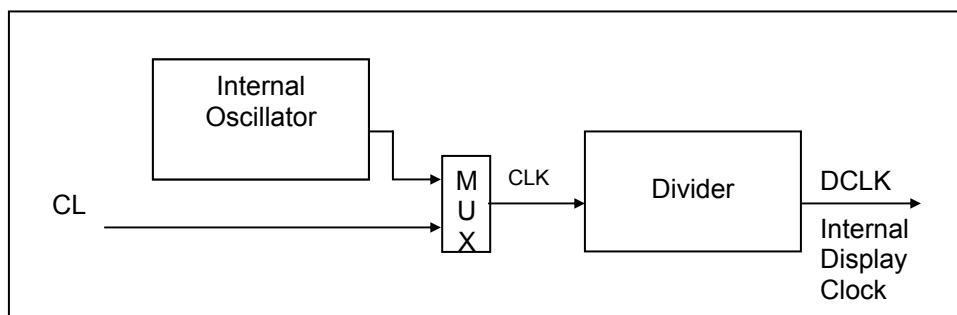


Figure 9 - Oscillator Circuit

This module is an On-Chip low power RC oscillator circuitry (Figure 9). The oscillator generates the clock for the Display Timing Generator.

Current Control and Voltage Control

This block is used to derive the incoming power sources into the different levels of internal use voltage and current. V_{CC} and V_{DD} are external power supplies. V_{REF} is reference voltage, which is used to derive driving voltage for segments and commons. I_{REF} is a reference current source for segment current drivers.

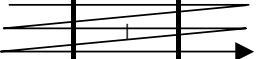
Segment Drivers/Common Drivers

Segment drivers deliver 128 current sources to drive OLED panel. The driving current can be adjusted from 0 to 300uA with 128 steps. Common drivers generate voltage scanning pulse.

Graphic Display Data RAM (GDDRAM)

The GDDRAM is a bit mapped static RAM holding the bit pattern to be displayed. The size of the RAM is 128x80x4 bits. For mechanical flexibility, re-mapping on both Segment and Common outputs can be selected by software. (Refer to Table 3-7 for GDDRAM address map description)

Table 9 – GDDRAM address map showing Horizontal Address Increment $A[2]=0$, Column Address Re-map $A[0]=0$, Nibble Re-map $A[1]=0$, COM Re-map $A[4]=0$, and Display Start Line=00H (Data byte sequence: D0, D1, ... , D5118, D5119)

		SEG0	SEG1	SEG2	SEG3		SEG124	SEG125	SEG126	SEG127	SEG Outputs Column Address (HEX)
		00		01			3E		3F		
COM0	00	D0[3:0]	D0[7:4]	D1[3:0]	D1[7:4]		D62[3:0]	D62[7:4]	D63[3:0]	D63[7:4]	
COM1	01	D64[3:0]	D64[7:4]	D65[3:0]	D65[7:4]		D126[3:0]	D126[7:4]	D127[3:0]	D127[7:4]	
											
COM78	4E	D4992[3:0]	D4992[7:4]	D4993[3:0]	D4993[7:4]		D5054[3:0]	D5054[7:4]	D5055[3:0]	D5055[7:4]	
COM79	4F	D5056[3:0]	D5056[7:4]	D5057[3:0]	D5057[7:4]		D5118[3:0]	D5118[7:4]	D5119[3:0]	D5119[7:4]	

COM Outputs

Row Address (HEX)

(Display Startline=0)

Table 10 – GDDRAM address map showing Horizontal Address Increment A[2]=1, Column Address Re-map A[0]=0, Nibble Re-map A[1]=0, COM Re-map A[4]=0, and Display Start Line=00H (Data byte sequence: D0, D1, ... , D5118, D5119)

		SEG0	SEG1	SEG2	SEG3		SEG124	SEG125	SEG126	SEG127	SEG Outputs
		00		01			3E		3F		Column Address
COM0	00	D0[3:0]	D0[7:4]	D80[3:0]	D80[7:4]		D4960[3:0]	D4960[7:4]	D5040[3:0]	D5040[7:4]	(HEX)
COM1	01	D1[3:0]	D1[7:4]	D81[3:0]	D81[7:4]		D4961[3:0]	D4961[7:4]	D5041[3:0]	D5041[7:4]	
COM78	4E	D78[3:0]	D78[7:4]	D158[3:0]	D158[7:4]		D5038[3:0]	D5038[7:4]	D5118[3:0]	D5118[7:4]	
COM79	4F	D79[3:0]	D79[7:4]	D159[3:0]	D159[7:4]		D5039[3:0]	D5039[7:4]	D5119[3:0]	D5119[7:4]	
COM Outputs	Row Address (HEX)										

(Display Startline=0)

Table 11 – GDDRAM address map showing Horizontal Address Increment A[2]=0, Column Address Re-map A[0]=1, Nibble Re-map A[1]=1, COM Re-map A[4]=0, and Display Start Line=00H (Data byte sequence: D0, D1, ... , D5118, D5119)

		SEG0	SEG1	SEG2	SEG3		SEG124	SEG125	SEG126	SEG127	SEG Outputs	
		3F		3E			01		00		Column Address	
COM0	00	D63[7:4]	D63[3:0]	D62[7:4]	D62[3:0]		D1[7:4]	D1[3:0]	D0[7:4]	D0[3:0]	(HEX)	
COM1	01	D127[7:4]	D127[3:0]	D126[7:4]	D126[3:0]		D65[7:4]	D65[3:0]	D64[7:4]	D64[3:0]		
COM78	4E	D5055[7:4]	D5055[3:0]	D5054[7:4]	D5054[3:0]		D4993[7:4]	D4993[3:0]	D4992[7:4]	D4992[3:0]		
COM79	4F	D5119[7:4]	D5119[3:0]	D5118[7:4]	D5118[3:0]		D5057[7:4]	D5057[3:0]	D5056[7:4]	D5056[3:0]		
COM Outputs	Row Address (HEX)											

(Display Startline=0)

Table 12 – GDDRAM address map showing Horizontal Address Increment A[2]=0, Column Address Re-map A[0]=0, Nibble

		SEG0	SEG1	SEG2	SEG3		SEG124	SEG125	SEG126	SEG127	SEG Outputs
		00		01			3E		3F		Column Address
COM15	0F	D0[3:0]	D0[7:4]	D1[3:0]	D1[7:4]		D62[3:0]	D62[7:4]	D63[3:0]	D63[7:4]	(HEX)
COM14	0E	D64[3:0]	D64[7:4]	D65[3:0]	D65[7:4]		D126[3:0]	D126[7:4]	D127[3:0]	D127[7:4]	
COM17	11	D4992[3:0]	D4992[7:4]	D4993[3:0]	D4993[7:4]		D5054[3:0]	D5054[7:4]	D5055[3:0]	D5055[7:4]	
COM16	10	D5056[3:0]	D5056[7:4]	D5057[3:0]	D5057[7:4]		D5118[3:0]	D5118[7:4]	D5119[3:0]	D5119[7:4]	
COM Outputs	Row Address (HEX)										

(Display Startline=10H)

Re-map A[1]=0, COM Re-map A[4]=1, and Display Start Line=16H (Data byte sequence: D0, D1, ... , D5118, D5119)

Table 13 – GDDRAM address map showing Horizontal Address Increment A[2]=0, Column Address Re-map A[0]=0, Nibble Re-map A[1]=0, COM Re-map A[4]=0, Display Start Line=00H (Data byte sequence: D0, D1, ... , D4834, D4835), Column Start Address=01H, Column End Address=3EH, Row Start Address=01H and Row End Address=4EH

		SEG0	SEG1	SEG2	SEG3		SEG124	SEG125	SEG126	SEG127	SEG Outputs Column Address (HEX)	
		00		01			3E		3F			
COM0	00											
COM1	01			D0[3:0]	D0[7:4]		D61[3:0]	D61[7:4]				
												
COM78	4E			D4774[3:0]	D4774[7:4]		D4835[3:0]	D4835[7:4]				
COM79	4F											
COM Outputs	Row Address (HEX)											

(Display Startline=0)

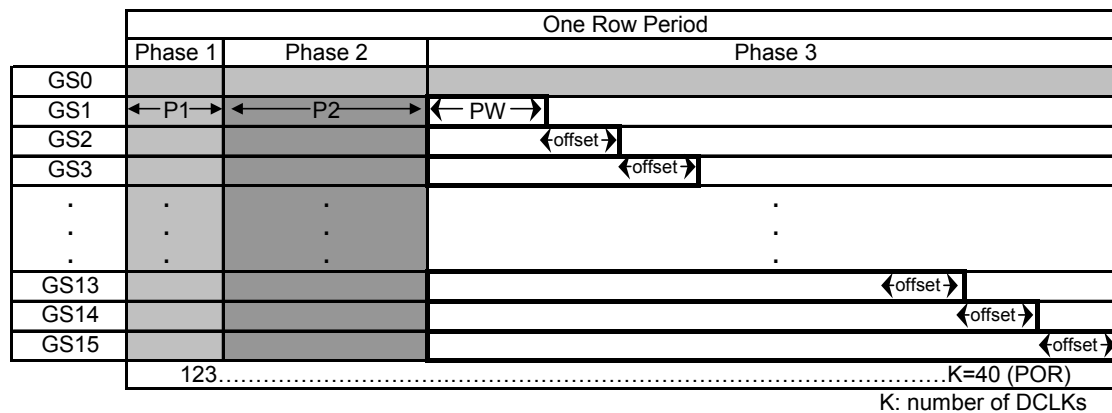
Gray Scale Decoder

There are 16 gray levels from GS0 to GS15. The gray scale of the display is defined by the pulse width (PW) of current drive phase, GS0 has no pre-charge and no current drive. Each L value represents an offset to the corresponding gray scale level. See below table and graphical representation:

Table14 - Gray scale pulse width set table

	Description	Number of DCLKs
L1	Set GS1 level Pulse Width	0-7
L2	Set GS2 level Pulse Width Offset	1-8
L3	Set GS3 level Pulse Width Offset	1-8
.	.	.
.	.	.
.	.	.
L13	Set GS13 level Pulse Width Offset	1-8
L14	Set GS14 level Pulse Width Offset	1-8
L15	Set GS15 level Pulse Width Offset	1-8

DCLK: Internal Display Clock. It is used for defining phase clock period.



no precharge and current drive

Precharge

Current Drive

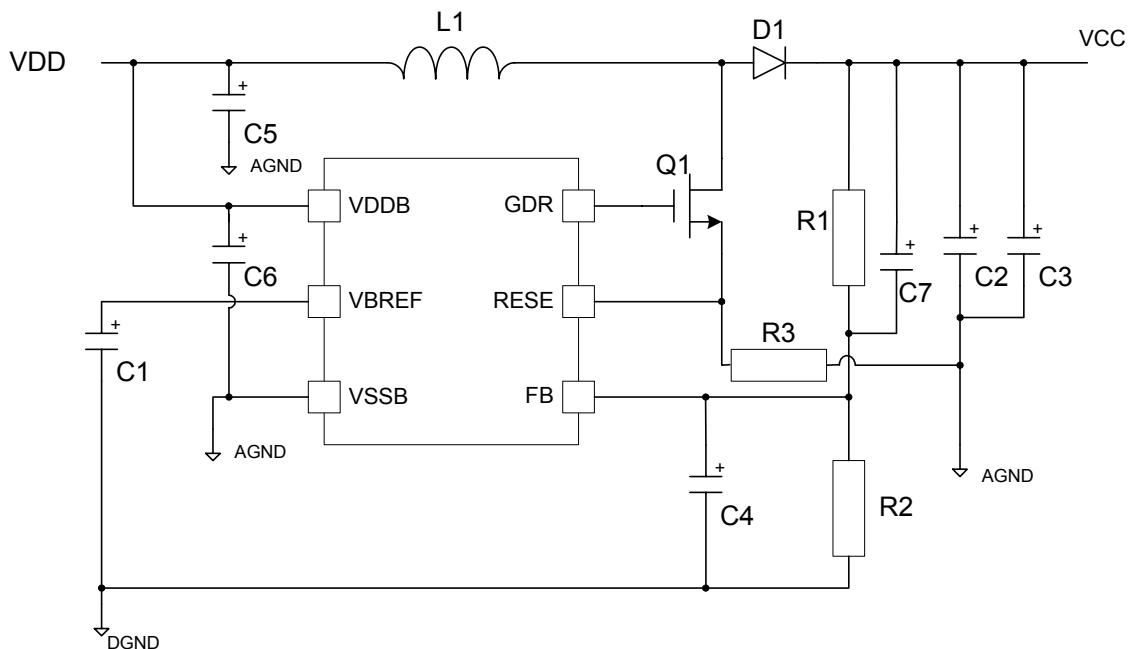
Figure 10 - Gray scale pulse width set diagram

Table 15 - Gray scale pulse width default values

RESE T	Result
L1=1	GS1 level Pulse width=1
L2=1	GS2 level Pulse width=3
L3=1	GS3 level Pulse width=5
L4=1	GS4 level Pulse width=7
L5=1	GS5 level Pulse width=9
L6=1	GS6 level Pulse width=11
L7=1	GS7 level Pulse width=13
L8=1	GS8 level Pulse width=15
L9=1	GS9 level Pulse width=17
L10=1	GS10 level Pulse width=19
L11=1	GS11 level Pulse width=21
L12=1	GS12 level Pulse width=23
L13=1	GS13 level Pulse width=25
L14=1	GS14 level Pulse width=27
L15=1	GS15 level Pulse width=29

DC-DC Voltage Converter

It is a switching voltage generator circuit, designed for handheld applications. In SSD0323, internal DC-DC voltage converter accompanying with an external application circuit (shown in below figure) can generate a high voltage supply V_{CC} from a low voltage supply input V_{DD} . V_{CC} is the voltage supply to the OLED driver block. Below application circuit is an example for the input voltage of 3V V_{DD} to generate V_{CC} of 12V @20mA ~ 30mA application.



Remark:

1. L1, D1, Q1, C5 should be grouped closed together on PCB layout.
2. R1, R2, C1, C4 should be grouped closed together on PCB layout.
3. The VCC output voltage level can be adjusted by R1 and R2, the reference formula is:

$$V_{CC} = 1.2 \times (R1 + R2) / R2$$
The value of (R1+R2) should be between 500k to 1M Ohm.

Table 16 - Passive component selection:

Components	Typical Value	Remark
L1	Inductor, 22μH	2A
D1	Schottky diode	2A, 25V e.g. 1N5822
Q1	MOSFET	N-FET with low $R_{DS(on)}$ and low V_{th} voltage. e.g. MGSF1N02LT1 [ON SEMICONDUCTOR]
R1, R2	Resistor	1%, 1/10W
R3	Resistor, 1.5Ω	1%, 1/2W
C1	Capacitor, 1μF	16V
C2	Capacitor, 22μF	Low ESR, 25V
C3	Capacitor, 1μF	16V
C4	Capacitor, 10nF	16V
C5	Capacitor, 1 ~ 10 μF	16V
C6	Capacitor, 0.1 ~ 1μF	16V
C7	Capacitor, 15nF	16V

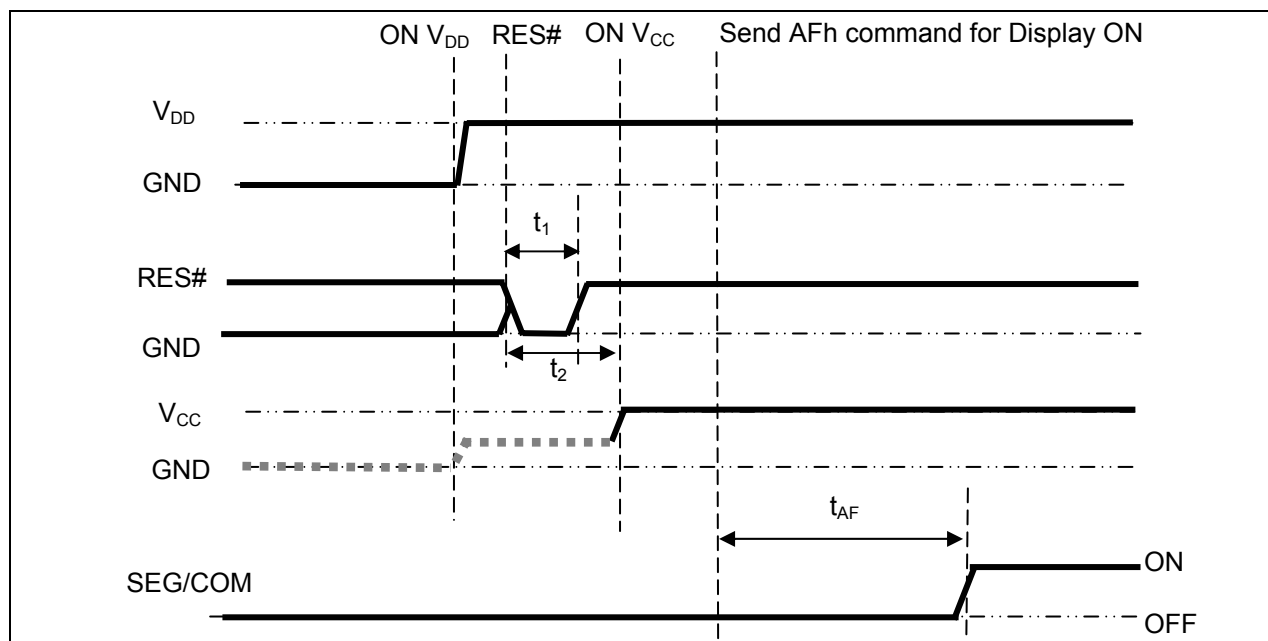
Power ON and OFF sequence

The following figures illustrate the recommended power ON and power OFF sequence of SSD0323.

Power ON sequence:

1. Power ON V_{DD} .
2. After V_{DD} become stable, set RES# pin LOW (logic LOW) for at least 3us (t_1) and then HIGH (logic HIGH).
3. After set RES# pin LOW (logic LOW), wait for at least 3us (t_2). Then Power ON V_{CC} .⁽¹⁾
4. After V_{CC} become stable, send command AFh for display ON. SEG/COM will be ON after 100ms (t_{AF}).

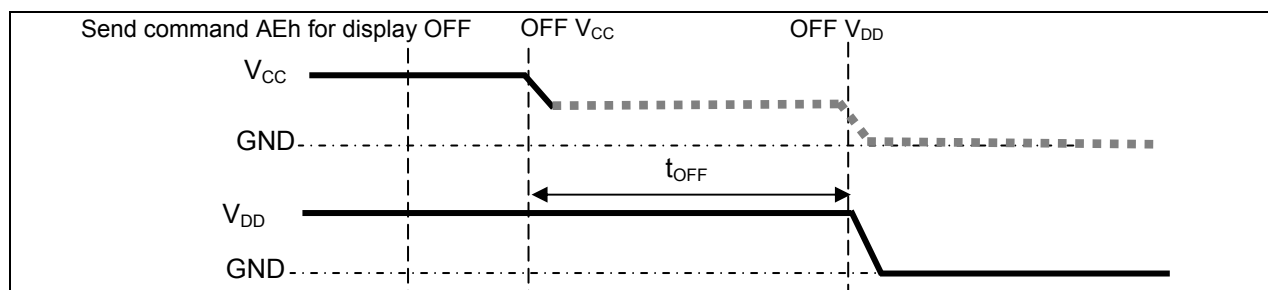
Figure 11 - The Power ON sequence



Power OFF sequence:

1. Send command AEh for display OFF.
2. Wait until panel discharges completely.
3. Power OFF V_{CC} .^{(1), (2)}
4. Wait for t_{OFF} . Power OFF V_{DD} . (where Minimum $t_{OFF}=0ms$, Typical $t_{OFF}=100ms$)

Figure 12 - The Power OFF sequence



Note:

⁽¹⁾ Since an ESD protection circuit is connected between V_{DD} and V_{CC} , V_{CC} becomes lower than V_{DD} whenever V_{DD} is ON and V_{CC} is OFF as shown in the dotted line of V_{CC} in Figure 11 and Figure 12.

⁽²⁾ V_{CC} should be kept float when it is OFF.

COMMAND TABLE

Table 17 - Command Table

(D/C# =0, R/W#(WR#)=0, E (RD#)=1)

D/C	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0 0 0	15 A[5:0] B[5:0]	0 * *	0 * *	0 A ₅ B ₅	1 A ₄ B ₄	0 A ₃ B ₃	1 A ₂ B ₂	0 A ₁ B ₁	1 A ₀ B ₀	Set Column Address	Second command A[5:0] sets the column start address from 0-63, RESET = 00H. Third command B[5:0] sets the column end address from 0-63, RESET = 3FH.
0 0 0	75 A[6:0] B[6:0]	0 * *	1 A ₆ B ₆	1 A ₅ B ₅	1 A ₄ B ₄	0 A ₃ B ₃	1 A ₂ B ₂	0 A ₁ B ₁	1 A ₀ B ₀	Set Row address	Second command A[6:0]sets the row start address from 0-79, RESET = 00H. Third command B[6:0] sets the row end address from 0-79, RESET = 4FH.
0 0	81 A[6:0]	1 *	0 A ₆	0 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Contrast Control Register	Double byte command to select 1 out of 128 contrast steps. Contrast increases as level increase. The level is set to 40H after RESET
0	84~86	1	0	0	0	0	1	X ₁	X ₀	Set Current Range	84H = Quarter Current Range (RESET) 85H = Half Current Range 86H = Full Current Range
0 0	A0 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	0 A ₀	Set Re-map	A[0]=0, Disable Column Address Re-map (RESET) A[0]=1, Enable Column Address Re-map A[1]=0, Disable Nibble Re-map (RESET) A[1]=1, Enable Nibble Re-map A[2]=0, Horizontal Address Increment (RESET) A[2]=1, Vertical Address Increment A[4]=0, Disable COM Re-map disable (RESET) A[4]=1, Enable COM Re-map A[5]=0, Reserved (RESET) A[5]=1, Reserved A[6]=0, Disable COM Split Odd Even (RESET) A[6]=1, Enable COM Split Odd Even
0 0	A1 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	0 A ₃	0 A ₂	0 A ₁	1 A ₀	Set Display Start Line	Set display RAM display start line register from 0-79. Display start line register is reset to 00H after RESET.
0 0	A2 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	0 A ₃	0 A ₂	1 A ₁	0 A ₀	Set Display Offset	Set vertical scroll by COM from 0-79. The value is reset to 00H after RESET.
0	A4~A7	1	0	1	0	0	X ₂	X ₁	X ₀	Set Display Mode	A4H = Normal Display (RESET) A5H = Entire Display ON, all pixels turns ON in GS level 15 A6H = Entire Display OFF, all pixels turns OFF A7H = Inverse Display

D/C	Hex	D7	D6	D5	D4	D3	D2	D1	D0	Command	Description
0 0	A8 A[6:0]	1 *	0 A ₆	1 A ₅	0 A ₄	1 A ₃	0 A ₂	0 A ₁	0 A ₀	Set Multiplex Ratio	The next command determines multiplex ratio N from 16MUX-80MUX, RESET=4FH(80MUX)
0 0	AD A[1:0]	1 *	0 *	1 *	0 *	1 *	1 *	0 1	1 A ₀	Set Master Configuration	A[0] = 0, Disable DC-DC converter A[0] = 1, Enable DC-DC converter (RESET)
0	AE	1	0	1	0	1	1	1	0	Set Display OFF	AEH = Display OFF (Sleep mode) (RESET)
0	AF	1	0	1	0	1	1	1	1	Set Display ON	AFH = Display ON
0 0	B0 A[5:0]	1 *	0 *	1 A ₅	1 A ₄	0 A ₃	0 A ₂	0 A ₁	0 A ₀	Set Pre-charge Compensation Enable	A[5:0] = 08h (RESET) A[5:0] = 28h, Enable pre-charge compensation
0 0	B4 A[2:0]	1 *	0 *	1 *	1 *	0 *	1 A ₂	0 A ₁	0 A ₀	Set Pre-charge Compensation Level	A[2:0] = 0 (RESET) A[2:0] = 3h, Recommended level
0 0	BF A[3:0]	1 *	0 *	1 *	1 *	1 A ₃	1 A ₂	1 A ₁	1 A ₀	Set Segment Low Voltage(VSL)	Second command A[3:0] sets the VSL voltage as follow: 1000-1110 A[3:0] = 0010 connects to VSS A[3:0] = 1110 (RESET) (When VDD > 2.5V)
0 0	BE A[4:0]	1 *	0 *	1 0	1 A ₄	1 A ₃	1 A ₂	1 A ₁	0 A ₀	Set V _{COMH} Voltage	Second command A[4:0] sets the V _{COMH} voltage level A[4:0] : 00000 0.51*V _{REF} 00001 0.52* V _{REF} 11101 0.81* V _{REF} (RESET) 11110 0.82* V _{REF} 11111 0.84* V _{REF}
0 0	BC A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	1 A ₃	1 A ₂	0 A ₁	0 A ₀	Set Precharge Voltage	Second command A[7:0] sets the precharge voltage level A[7:0] 1xxxxxxx connects to V _{COMH} (RESET) 001xxxxx 1.0 * V _{REF} 00000000 0.51* V _{REF} 00000001 0.52* V _{REF} 00011111 0.84* V _{REF}
0 0 0	B1 A[3:0] A[7:4]	1 * A ₇	0 * A ₆	1 * A ₅	1 * A ₄	0 A ₃ *	0 A ₂ *	0 A ₁ *	1 A ₀ *	Set Phase Length	A[3:0] = P1, phase 1 period of 1-15 DCLKs, RESET = 3DCLKS = 3H A[7:4] = P2, phase 2 period of 1-15 DCLKs, RESET = 5DCLKS = 5H Note (1) 0 DCLK is invalid in phase 1 & phase 2
0 0	B2 A[7:0]	1 A ₇	0 A ₆	1 A ₅	1 A ₄	0 A ₃	0 A ₂	1 A ₁	0 A ₀	Set Row Period	The next command sets the number of DCLKs, K, per row between 2-158DLKS, RESET = 37DLKS = 25H The K value should be set as K = P1+P2+GS15 pulse width (RESET: 3+5+29DLKS)

Table 18: Graphic acceleration command

Set (GAC) (D/C# = 0, R/W#(WR#)= 0, E(RD#) = 1) unless specific setting is stated

Graphic acceleration command										
D/C#	Hex	D7	D6	D5	D4	D3	D2	D2	D0	Command
0	23	0	0	1	0	0	0	1	1	Graphic Acceleration Command Options
0	A[4:0]	*	*	*	A ₄	*	*	A ₁	A ₀	
										A[0] = 0b: Disable Fill rectangle A[0] = 1b: Enable Fill rectangle (RESET) A[1] = 0b: Disable x-wrap(RESET) A[1] = 1b: Enable wrap around in x-direction during copying and scrolling A[4] = 0b: Disable reverse copy (RESET) A[4] = 1b: Enable reverse during copying.
0	24	0	0	1	0	0	1	0	0	Draw Rectangle
0	A[5:0]	*	*	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	B[6:0]	*	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	
0	C[5:0]	*	*	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀	
0	D[6:0]	*	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	
0	E[7:0]	E ₇	E ₆	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀	
										A[5:0]: Column Address of Start B[6:0]: Row Address of Start C[5:0]: Column Address of End D[6:0]: Row Address of End E[7:0]: Set Gray scale pattern E[7:0] This byte is divided into two nibbles. The most significant 4 bits represent the gray scale level of the left pixel of each group. The least significant 4 bits represent the gray scale level of the right pixel of each group. Please refer to Figure 17 for the gray scale pattern setting examples.
										Note: ⁽¹⁾ 0 ≤ A < C ≤ 63 ⁽²⁾ 0 ≤ B < D ≤ 79
0	25	0	0	1	0	0	1	0	1	Copy
0	A[5:0]	*	*	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	B[6:0]	*	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	
0	C[5:0]	*	*	C ₅	C ₄	C ₃	C ₂	C ₁	C ₀	
0	D[6:0]	*	D ₆	D ₅	D ₄	D ₃	D ₂	D ₁	D ₀	
0	E[5:0]	*	*	E ₅	E ₄	E ₃	E ₂	E ₁	E ₀	
0	F[6:0]	*	F ₆	F ₅	F ₄	F ₃	F ₂	F ₁	F ₀	
										A[5:0]: Column Address of Start B[6:0]: Row Address of Start C[5:0]: Column Address of End D[6:0]: Row Address of End E[5:0]: Column Address of New Start F[6:0]: Row Address of New Start
										Note: ⁽¹⁾ 0 ≤ A < C ≤ 63 ⁽²⁾ 0 ≤ B < D ≤ 79 ⁽³⁾ 0 ≤ E ≤ 63 ⁽⁴⁾ 0 ≤ F ≤ 79

Graphic acceleration command										
D/C#	Hex	D7	D6	D5	D4	D3	D2	D2	D0	Description
0	26	0	0	1	0	0	1	1	0	Horizontal Scroll Note: ⁽¹⁾ Scrolling operates during display ON. ⁽²⁾ The parameters should not be changed after scrolling is activated
0	A[5:0]	*	*	A ₅	A ₄	A ₃	A ₂	A ₁	A ₀	
0	B[6:0]	*	B ₆	B ₅	B ₄	B ₃	B ₂	B ₁	B ₀	
0	C[1:0]	*	*	*	*	*	*	C ₁	C ₀	
0	2E	0	0	1	0	1	1	1	0	This command deactivates the scrolling action. Note ⁽¹⁾ After sending 2Eh command to deactivate the scrolling action, the ram data needs to be rewritten.
0	2F	0	0	1	0	1	1	1	1	This command activates the scrolling function according to the setting done by Horizontal Scroll command 26h. Note ⁽¹⁾ The “wrap around in x-direction” function must be enabled before scrolling start. i.e. Bit A{1} of command 23h must be set to 1b before issuing 2F command.

Table 19 - Read Command Table

(D/C#=0, R/W#(WR#)=1, E(RD#)=1 for 6800 or E(RD#)=0 for 8080)

D ₇ D ₆ D ₅ D ₄ D ₃ D ₂ D ₁ D ₀	Status Register Read	D7 = 0:reserved D7 = 1:reserved D6 = 0:indicates the display is ON D6 = 1:indicated the display is OFF D5 = 0:reserved D5 = 1:reserved D4 = 0:reserved D4 = 1:reserved
---	----------------------	---

Note: Patterns other than that given in Command Table are prohibited to enter to the chip as a command; Otherwise, unexpected result will occur

Data Read / Write

To read data from the GDDRAM, input HIGH to R/W# (WR#) pin and D/C# pin for 6800-series parallel mode, LOW to E (RD#) pin and HIGH to D/C# pin for 8080-series parallel mode.

In horizontal address increment mode, GDDRAM column address pointer will be increased by one automatically after each data read. In vertical address increment mode, GDDRAM row address pointer will be increased by one automatically after each data read.

Also, a dummy read is required before the first data read. See Figure 4 and Figure 7 in Functional Description.

To write data to the GDDRAM, input LOW to R/W#(WR#) pin and HIGH to D/C# pin for 6800-series parallel mode and 8080-series parallel mode. For serial interface mode, it is always in write mode. In horizontal address increment mode, GDDRAM column address pointer will be increased by one automatically after each data write. In vertical address increment mode, GDDRAM row address pointer will be increased by one automatically after each data write.

It should be noted that, in horizontal address increment mode, the row address pointer would be increased by one automatically if the column address pointer wraps around. In vertical address increment mode, the column address pointer will be increased by one automatically if the row address pointer wraps around.

Table 20 - Address Increment Table (Automatic)

D/C#	R/W# (WR#)	Comment	Address Increment
0	0	Write Command	No
0	1	Read Status	No
1	0	Write Data	Yes
1	1	Read Data	Yes

COMMAND DESCRIPTIONS

Set Column Address

This command specifies column start address and end address of the display data RAM. This command also sets the column address pointer to column start address.

Set Row Address

This command specifies row start address and end address of the display data RAM. This command also sets the row address pointer to row start address.

Set Contrast Control Register

This command is to set Contrast Setting of the display. The chip has 128 contrast steps from 00H to 7FH. The segment output current increases with the increase of contrast step. See Figure 13 below.

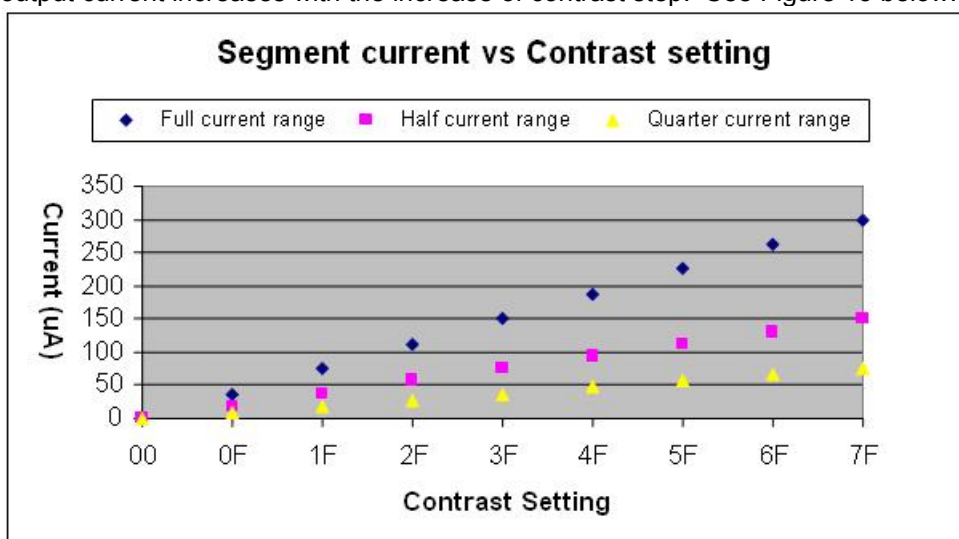


Figure 13 - Segment current vs Contrast setting

Set Current Range

This command is used to select quarter range or half range or full range current mode. With the same contrast level, quarter range mode will give a quarter of the current output of the full range mode. Similar to half range current mode, it will give a half of the current output of the full range mode. See Figure 13. In RESET, quarter range current mode is default.

Set Re-map

This command changes the mapping between the display data column address and segment driver, row address and common driver. It allows flexibility in layout during OLED module assembly. See the Re-map setting below:

Column Address Re-map

If column address re-map is set, Col 0-63 map to SEG127-0, regardless of start column and end column commands.

Nibble Re-map

If nibble re-map is set, the two nibbles of the data bus for RAM access are re-mapped, such that (D7, D6, D5, D4, D3, D2, D1, D0) acts like (D3, D2, D1, D0, D7, D6, D5, D4)

This feature working with Column Address Re-map would produce an effect of flipping outputs SEG0-127 to SEG127-SEG0.

Address Increment Mode

If horizontal increment mode is set, the column address pointer advances after each RAM access.

If vertical increment mode is set, the row address pointer advances after each RAM access.

COM Re-map

If COM re-map is set, ROW 0-79 map to COM79-0, regardless of start and end row commands.

Set Display Start Line

This command is to set Display Start Line register to determine starting address of display RAM to be displayed by selecting a value from 0 to 79.

Set Display Offset

This is a double byte command. The next command specifies the mapping of display start line (it is assumed that COM0 is the display start line, display start line register equals to 0) to one of COM0-79.

For example, to move the COMX towards the COM0 direction for L lines, the 7-bit data in the second command should be given by L. In other words, to move the COMX towards the COM79 direction for L lines, the 7-bit data in the second command should be given by 80-L.

Set Display Mode

This command is used to set Normal Display, Entire Display ON, Entire Display OFF and Inverse Display. Set Entire Display ON forces the entire display to be at gray level "GS15" regardless of the contents of the display data RAM. Set Entire Display OFF forces the entire display to be at gray level "GS0" regardless of the contents of the display data RAM. Normal Display will turn the data to ON at the corresponding gray level. Inverse display will turn the data as follow:

Table 21 - Mapping of data with each gray scale level at different display mode

Data D _{x4} D _{x3} D _{x2} D _{x1}	Normal Display (A4H)	Entire Display ON (A5H)	Entire Display OFF (A6H)	Inverse Display (A7H)
0000	GS0	GS15	GS0	GS15
0001	GS1	GS15	GS0	GS14
0010	GS2	GS15	GS0	GS13
0011	GS3	GS15	GS0	GS12
0100	GS4	GS15	GS0	GS11
0101	GS5	GS15	GS0	GS10
0110	GS6	GS15	GS0	GS9
0111	GS7	GS15	GS0	GS8
1000	GS8	GS15	GS0	GS7
1001	GS9	GS15	GS0	GS6
1010	GS10	GS15	GS0	GS5
1011	GS11	GS15	GS0	GS4
1100	GS12	GS15	GS0	GS3
1101	GS13	GS15	GS0	GS2
1110	GS14	GS15	GS0	GS1
1111	GS15	GS15	GS0	GS0

Set Multiplex Ratio

This command sets multiplex ratio from 16 to 80. In RESET, multiplex ratio is 80.

Set DC-DC Converter

This command is used to enable or disable the internal DC-DC voltage converter. This command will be executed when display is ON.

Set Display ON/OFF

This command turns the display ON or OFF. When the display is OFF, the segment and common output are in high impedance state.

Set Segment Low voltage

This command is used to set segment low voltage (VSL). The value of VSL is the same for display all ON, display all OFF pattern with either internal or external DC-DC voltage converter.

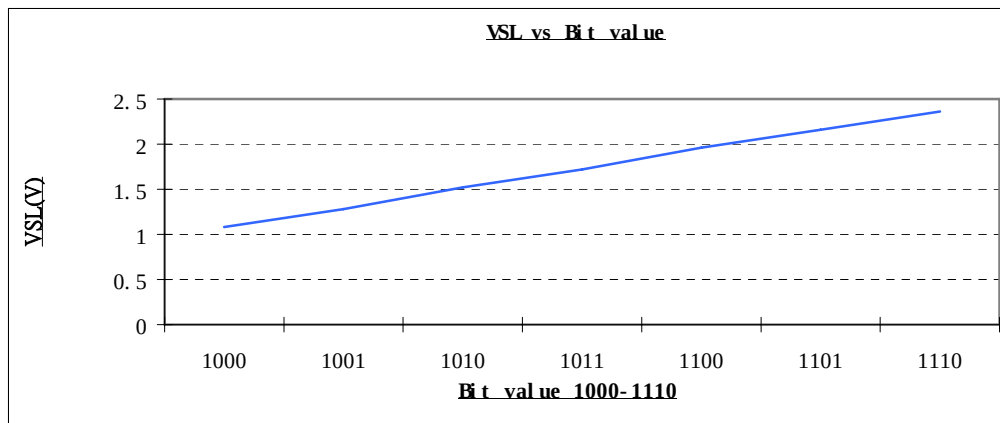


Figure 14 - VSL vs Bit value

Set V_{COMH} Voltage

This command is used to set V_{COMH} voltage level

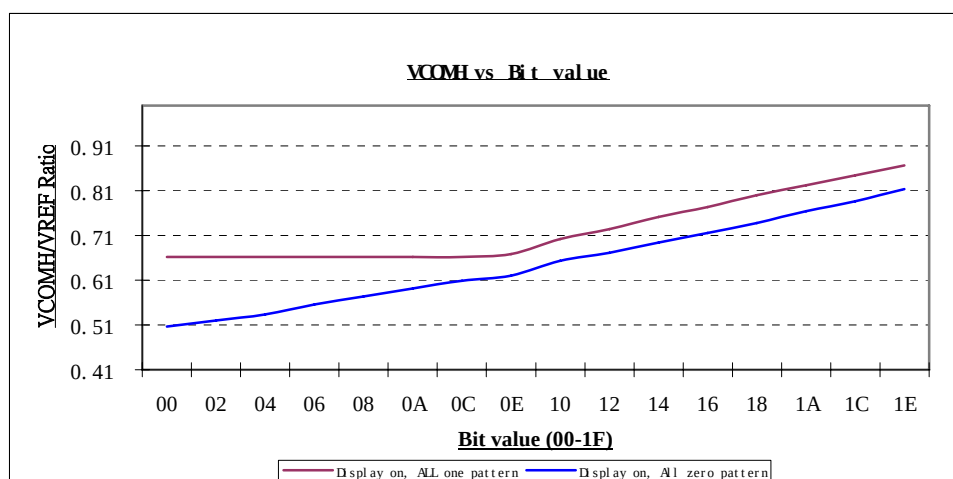


Figure 15 - VCOMH vs Bit value

Set Precharge Voltage

This command is used to set Precharge Voltage level

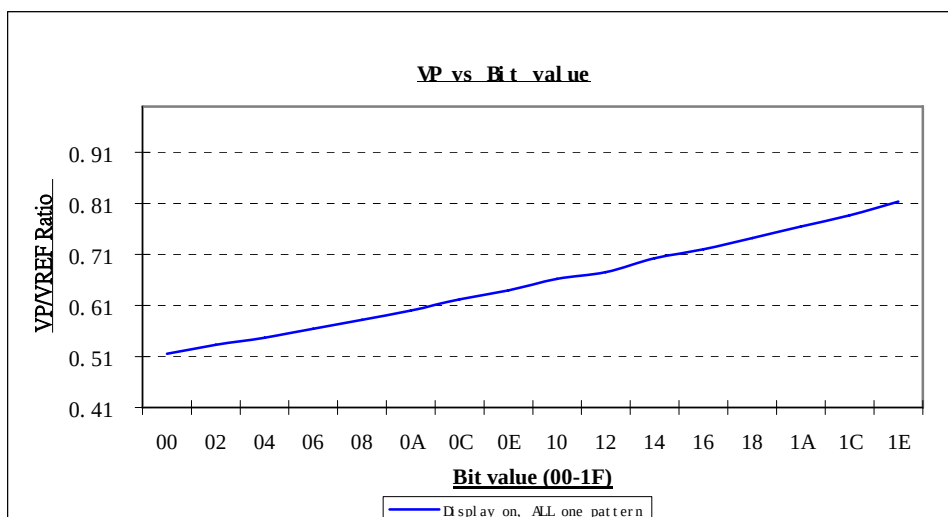


Figure 16 - VP vs Bit value

Set Phase Length

This is a double byte command. The lower nibble of the second byte selects phase 1 period (no pre-charge and current drive) from 1 to 16 DCLKs. RESET is A[3:0]=3d. The higher nibble of the second byte is used to select phase 2 period (pre-charge) from 1 to 16 DCLKs. RESET is A[7:4]=5d.

Set Row Period

This command is used to set the row period. It is defined by multiplying the internal display clock period by the number of internal display clocks per row (Value from 2-158d). RESET is 37d. The larger the value, the more precise of each gray scale level can be tuned. See "Gray Scale Table" command for details. Also, It is used to define the frame frequency with the use of "Display Clock Divide Ratio" command together.

Row period equals to the sum of phase 1, 2 periods and the pulse width of GS15. See equation in command table (B2H command)

Set Display Clock Divide Ratio

This command is used to set the frequency of the internal display clocks, DCLKs. It is defined by dividing the oscillator frequency by the divide ratio (Value from 1 to 16). RESET is 2. Frame frequency is determined by divide ratio, number of display clocks per row, MUX ratio and oscillator frequency. See equation in Table 24.

Set Oscillator Frequency

This is a double byte command. The lower nibble of the second byte is used to select the oscillator frequency. Default value is shown in Table 17.

Set Gray Scale Table

This command is used to set the gray scale table for the display. Except GS0, which has no pre-charge and current drive, each GS level is programmed by a set of offset values. As shown in Table 8, GS1 is defined with pulse width equals to the first offset value, L1, select from 0-7 internal display clocks. GS2 is defined with pulse width equals to GS1 plus the next offset value, L2, select from 1-8 internal display clocks. Similarly, the next GS level is defined with pulse width equals to its lower one GS level plus the next offset value, select from 1-8 internal display clocks. In normal operation, GS15 should take the full current drive period as its pulse width. Therefore, the row period should be set as the sum of phase 1 period, phase 2 period, and the pulse width of GS15 with the use of “Row period” command.

NOP

No Operation Command.

Status register Read

This command is issued by setting D/C# LOW during a data read (refer to Figure 20, Figure 21 and Figure 22 parallel interface waveform). It allows the MCU to monitor the internal status of the chip.

Graphic Acceleration Command Set Description

Graphic Acceleration Command Options (23h)

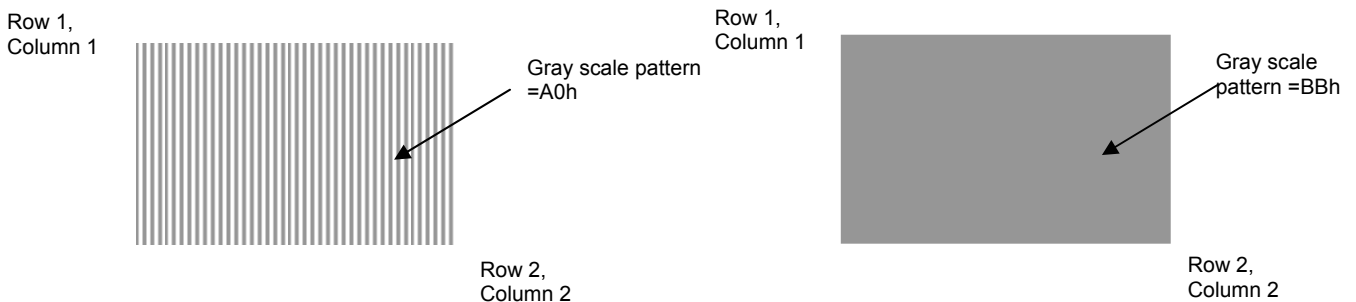
This command has two functions.

- Enable / Disable fill (A[0])
0 = Disable filling of rectangle in draw rectangle command.
1 = Enable filling of rectangle in draw rectangle command. (RESET)
- Enable / Disable x-warp (A[1])
0 = Disable wrap around in x-direction during copying and scrolling
1 = Enable wrap around in x-direction during copying and scrolling (RESET)
- Enable / Disable reverse copy (A[4])
0 = Disable reverse copy (RESET)
1 = During copy command, the new image colors are swapped such that "GS0" <-> "GS15", "GS1" <-> "GS14",

Draw Rectangle (24h)

Specify a starting point (Row 1, Column 1) and an ending point (Row 2, Column 2) as well as giving the desire gray scale pattern, a rectangle will then be drawn.

Figure 17 - Example of draw rectangle command



The following example illustrates the rectangle drawing command sequence.

1. Enter the "draw rectangle mode" by execute the command 24h
2. Set the starting column coordinates, Column 1. e.g., 01h.
3. Set the starting row coordinates, Row 1. e.g., 01h.
4. Set the finishing column coordinates, Column 9. e.g., 09h
5. Set the finishing row coordinates, Row 5. e.g., 05h
6. Set the gray scale pattern:

This byte is divided into two nibbles. The most significant 4 bits represent the gray scale level of the left pixel of each group. The least significant 4 bits represent the gray scale level of the right pixel of each group. Please refer to Figure 17 for the gray scale pattern setting examples.

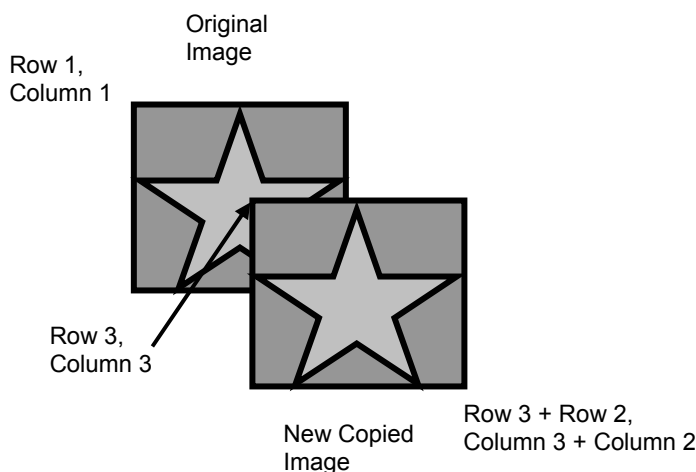
Copy (25h)

Copy the rectangular region defined by the starting point (Row 1, Column 1) and the ending point (Row 2, Column 2) to location (Row 3, Column 3). If the new coordinates are smaller than the ending points, the new image will overlap the original one.

The following example illustrates the copy procedure.

1. Enter the "copy mode" by execute the command 25h
2. Set the starting column coordinates, Column 1. E.g., 00h.
3. Set the starting row coordinates, Row 1. E.g., 00h.
4. Set the finishing column coordinates, Column 2. E.g., 05h
5. Set the finishing row coordinates, Row 2. E.g., 05h
6. Set the new column coordinates, Column 3. E.g., 03h
7. Set the new row coordinates, Row 3. E.g., 03h

Figure 18: Example of copy command

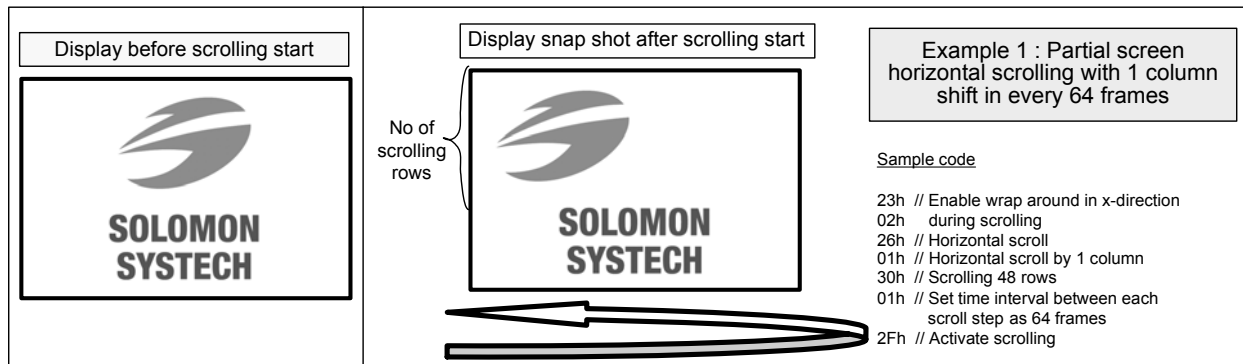


Horizontal Scroll (26h)

This command consists of 3 consecutive bytes to set up the scrolling parameters. It determines the horizontal scrolling offset, no. of scrolling rows and scrolling speed. Some scrolling examples are shown in Figure 19.

Before issuing this command, the scrolling must be deactivated (2Eh). Otherwise, RAM content may be corrupted.

Figure 19: Scrolling examples



Stop Moving (2Eh)

Stop motion of scrolling. After sending 2Eh command to deactivate the scrolling action, the ram data needs to be rewritten.

Start Moving (2Fh)

Start motion of scrolling. This command should only be issued after scrolling setup parameters are defined through command 26h and the function of wrap around in x-direction is enabled through 23h.

The following actions are prohibited after the horizontal scroll is activated

1. RAM access (Data write or read)
2. Changing scrolling setup parameters

MAXIMUM RATINGS

Table 22 - Maximum Ratings

(Voltage Reference to V_{SS})

Symbol	Parameter	Value	Unit
V_{DD}	Supply Voltage	-0.3 to +4	V
V_{CC}		0 to 17	V
V_{REF}		0 to 17	V
V_{SEG}	SEG output voltage	0 to V_{CC}	V
V_{COM}	COM output voltage	0 to $0.9 \times V_{CC}$	V
V_{in}	Input voltage	$V_{SS} - 0.3$ to $V_{DD} + 0.3$	V
T_A	Operating Temperature	-40 to +85	°C
T_{stg}	Storage Temperature Range	-65 to +150	°C

*Maximum Ratings are those values beyond which damage to the device may occur. Functional operation should be restricted to the limits in the Electrical Characteristics tables or Pin Description.

DC CHARACTERISTICS

Table 23 - DC Characteristics

(Unless otherwise specified, Voltage Referenced to V_{SS} , $V_{DD} = 2.4$ to $3.5V$, $T_A = 25^\circ C$)

Symbol	Parameter	Test Condition	Min	Typ	Max	Unit
V_{CC}	Operating Voltage	-	8	12	16	V
V_{DD}	Logic Supply Voltage	-	2.4	2.7	3.5	V
V_{OH}	HIGH Logic Output Level	$I_{OUT} = 100\mu A$, 3.3MHz	$0.9 \times V_{DD}$	-	V_{DD}	V
V_{OL}	LOW Logic Output Level	$I_{OUT} = 100\mu A$, 3.3MHz	0	-	$0.1 \times V_{DD}$	V
V_{IH}	HIGH Logic Input Level	-	$0.8 \times V_{DD}$	-	V_{DD}	V
V_{IL}	LOW Logic Input Level	-	0	-	$0.2 \times V_{DD}$	V
I_{SLEEP}	Sleep mode Current	No loading	-	0.2	5	uA
I_{CC}	V_{CC} Supply Current $V_{DD}=2.7V$, external $V_{CC}=12V$, $I_{REF}=10\mu A$, Frame rate=110Hz, All one pattern, Display ON, no loading	Contrast = 7F	-	700	-	uA
I_{DD}	V_{DD} Supply Current $V_{DD}=2.7V$, external $V_{CC}=12V$, $I_{REF}=10\mu A$, Frame rate=110Hz, All one pattern, Display ON, no loading	Contrast = 7F	-	-	650	uA
I_{SEG}	Segment Output Current $V_{DD}=2.7V$, $V_{CC}=12V$, $I_{REF}=10\mu A$, Frame rate=110Hz, Display ON, Segment pin under test is connected with a 20K resistive load to V_{SS}	Contrast = 7F	270	300	370	uA
		Contrast = 5F	-	225	-	
		Contrast = 3F	-	150	-	
		Contrast = 1F	-	75	-	
Dev	Segment output current uniformity $V_{DD}=2.7V$, $V_{CC}=12V$, $I_{REF}=10\mu A$, Contrast=7F	Adjacent pin	-	± 2	-	%
		Overall pin to pin	-3	-	+3	
V_{CC}	DC-DC converter output voltage	V_{DD} input=3V, L=22uH; R1=450Kohm; R2=50Kohm; $I_{CC} = 20mA$ (loading)	10	-	12	V
Pwr	DC-DC converter output power	V_{DD} input=3V, L=22uH; $V_{CC} = 12V$	-	-	400	mW

AC CHARACTERISTICS

Table 24 - AC Characteristics

(Unless otherwise specified, Voltage Referenced to V_{SS} , $V_{DD} = 2.4$ to $3.5V$, $T_A = 25^\circ C$.)

Symbol	Parameter	Test Condition	Min	Typ	Max	Unit
F_{OSC}	Oscillation Frequency of Display Timing Generator	$V_{DD} = 2.7V$	535	630	725	kHz
F_{FRM}	Frame Frequency for 128 MUX Mode	128x80 Graphic Display Mode, Display ON, Internal Oscillator Enabled	-	$F_{OSC} \times 1/(D \times K \times 80)$	-	Hz
RES#	Reset LOW pulse width	-	3	-	-	us
	Reset complete time	-	-	-	2	us

F_{OSC} stands for the frequency value of the internal oscillator and the value is measured when command B3h A[7:4] is in default value.

D: divide ratio

K: number of display clocks

Refer to command table for detail description

Table 25 - 6800-Series MPU Parallel Interface Timing Characteristics

($V_{DD} - V_{SS} = 2.4$ to $3.5V$, $T_A = 25^\circ C$)

Symbol	Parameter	Min	Typ	Max	Unit
t_{cycle}	Clock Cycle Time	300	-	-	ns
t_{AS}	Address Setup Time	0	-	-	ns
t_{AH}	Address Hold Time	0	-	-	ns
t_{DSW}	Write Data Setup Time	40	-	-	ns
t_{DHW}	Write Data Hold Time	15	-	-	ns
t_{DHR}	Read Data Hold Time	20	-	-	ns
t_{OH}	Output Disable Time	-	-	70	ns
t_{ACC}	Access Time	-	-	140	ns
PW_{CSL}	Chip Select Low Pulse Width (read) Chip Select Low Pulse Width (write)	120 60	-	-	ns
PW_{CSH}	Chip Select High Pulse Width (read) Chip Select High Pulse Width (write)	60 60	-	-	ns
t_R	Rise Time	-	-	15	ns
t_F	Fall Time	-	-	15	ns

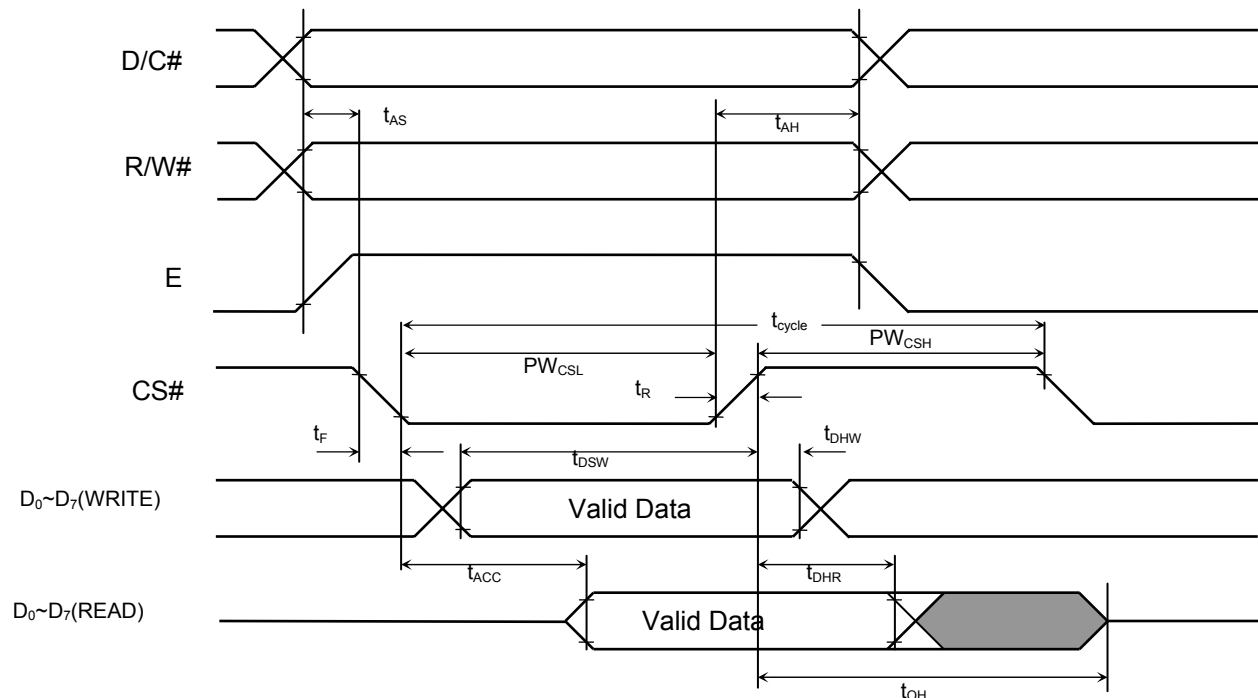


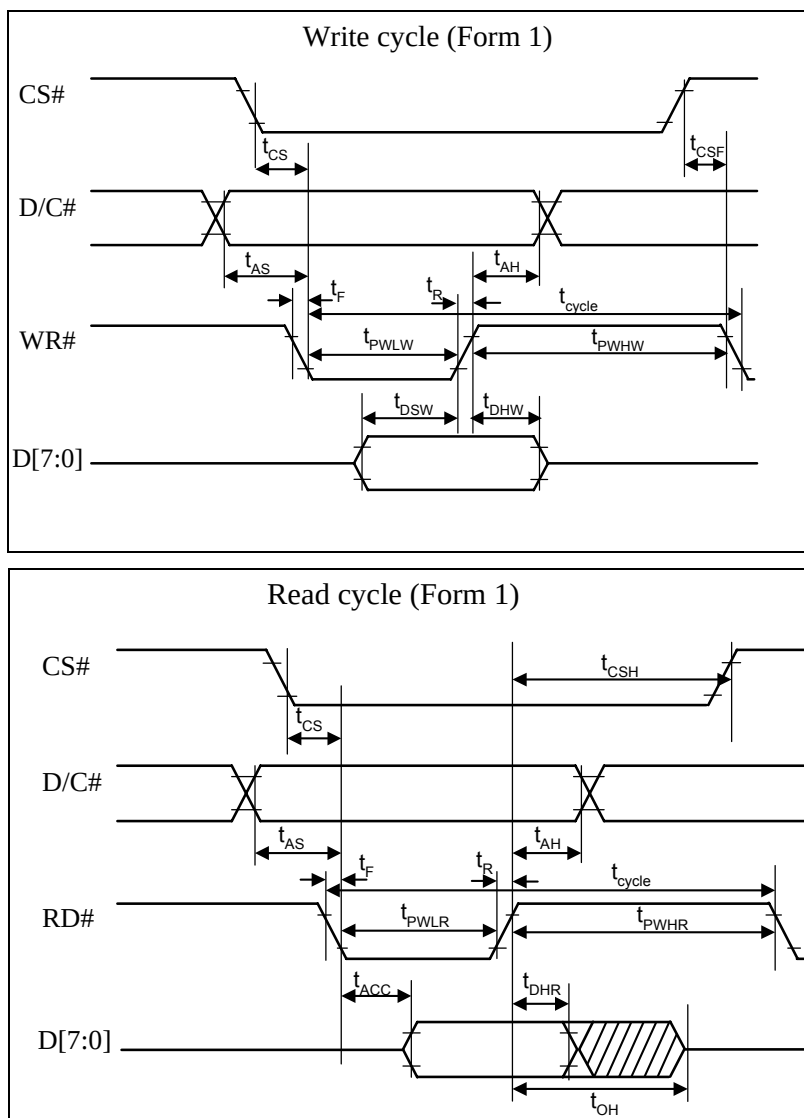
Figure 20 - 6800-series MPU Parallel Interface Characteristics

Table 26 - 8080-Series MPU Parallel Interface Timing Characteristics

($V_{DD} - V_{SS} = 2.4$ to $3.5V$, $T_A = 25^\circ C$)

Symbol	Parameter	Min	Typ	Max	Unit
t_{cycle}	Clock Cycle Time	300	-	-	ns
t_{AS}	Address Setup Time	10	-	-	ns
t_{AH}	Address Hold Time	0	-	-	ns
t_{DSW}	Write Data Setup Time	40	-	-	ns
t_{DHW}	Write Data Hold Time	15	-	-	ns
t_{DHR}	Read Data Hold Time	20	-	-	ns
t_{OH}	Output Disable Time	-	-	70	ns
t_{ACC}	Access Time	-	-	140	ns
t_{PWLR}	Read Low Time	120	-	-	ns
t_{PWLW}	Write Low Time	60	-	-	ns
t_{PWHR}	Read High Time	60	-	-	ns
t_{PWHW}	Write High Time	60	-	-	ns
t_R	Rise Time	-	-	15	ns
t_F	Fall Time	-	-	15	ns
t_{CS}	Chip select setup time	0	-	-	ns
t_{CSH}	Chip select hold time to read signal	0	-	-	ns
t_{CSF}	Chip select hold time	20	-	-	ns

Figure 21 - 8080-series parallel interface characteristics (Form 1)



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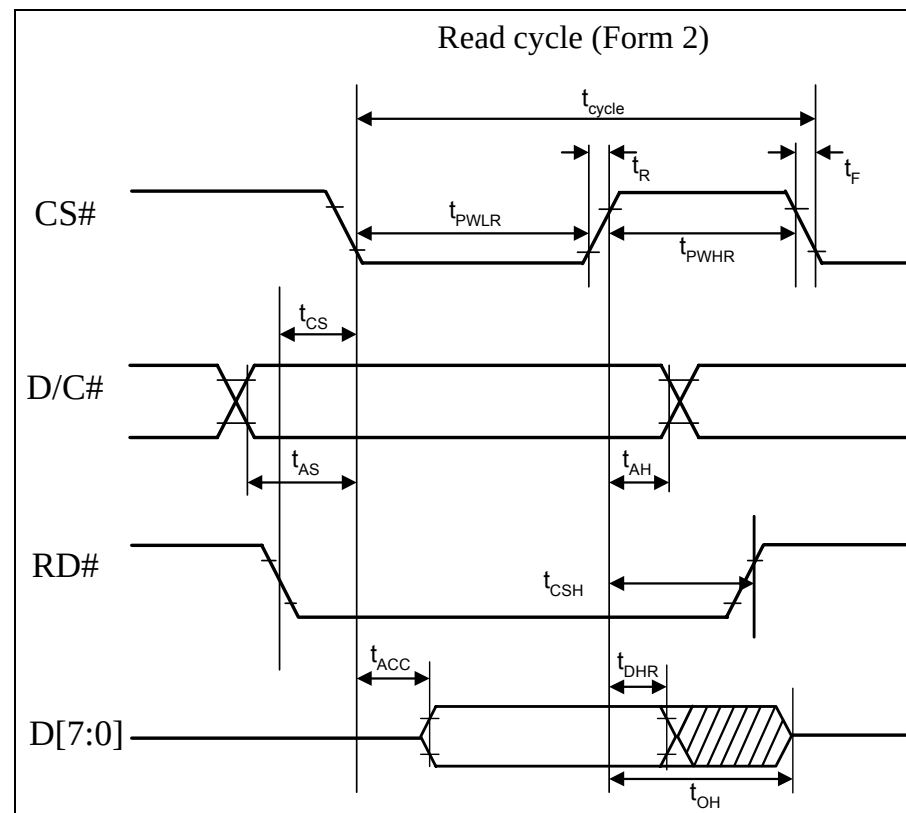


Table 27 - Serial Interface Timing Characteristics

($V_{DD} - V_{SS} = 2.4$ to $3.5V$, $T_A = 25^\circ C$)

Symbol	Parameter	Min	Typ	Max	Unit
t_{cycle}	Clock Cycle Time	250	-	-	ns
t_{AS}	Address Setup Time	150	-	-	ns
t_{AH}	Address Hold Time	150	-	-	ns
t_{CSS}	Chip Select Setup Time	120	-	-	ns
t_{CSH}	Chip Select Hold Time	60	-	-	ns
t_{DSW}	Write Data Setup Time	100	-	-	ns
t_{DHW}	Write Data Hold Time	100	-	-	ns
t_{CLKL}	Clock Low Time	100	-	-	ns
t_{CLKH}	Clock High Time	100	-	-	ns
t_R	Rise Time	-	-	15	ns
t_F	Fall Time	-	-	15	ns

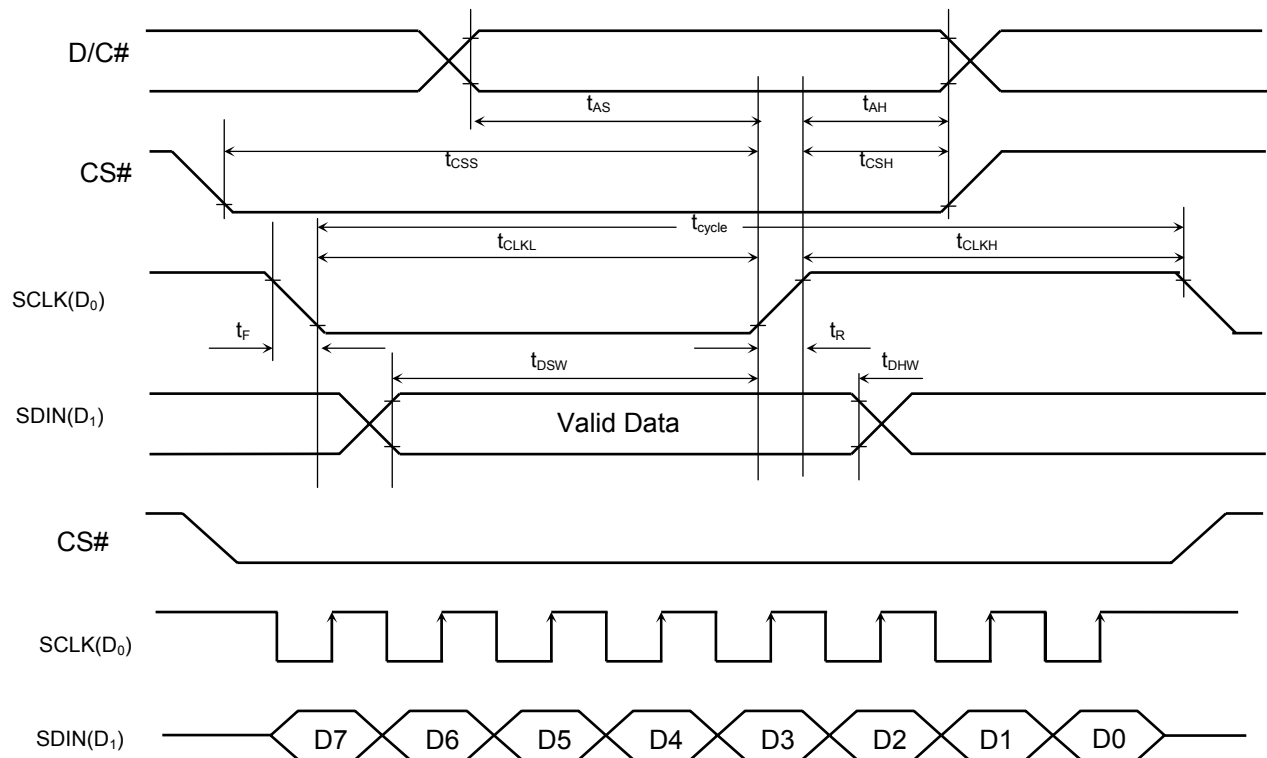
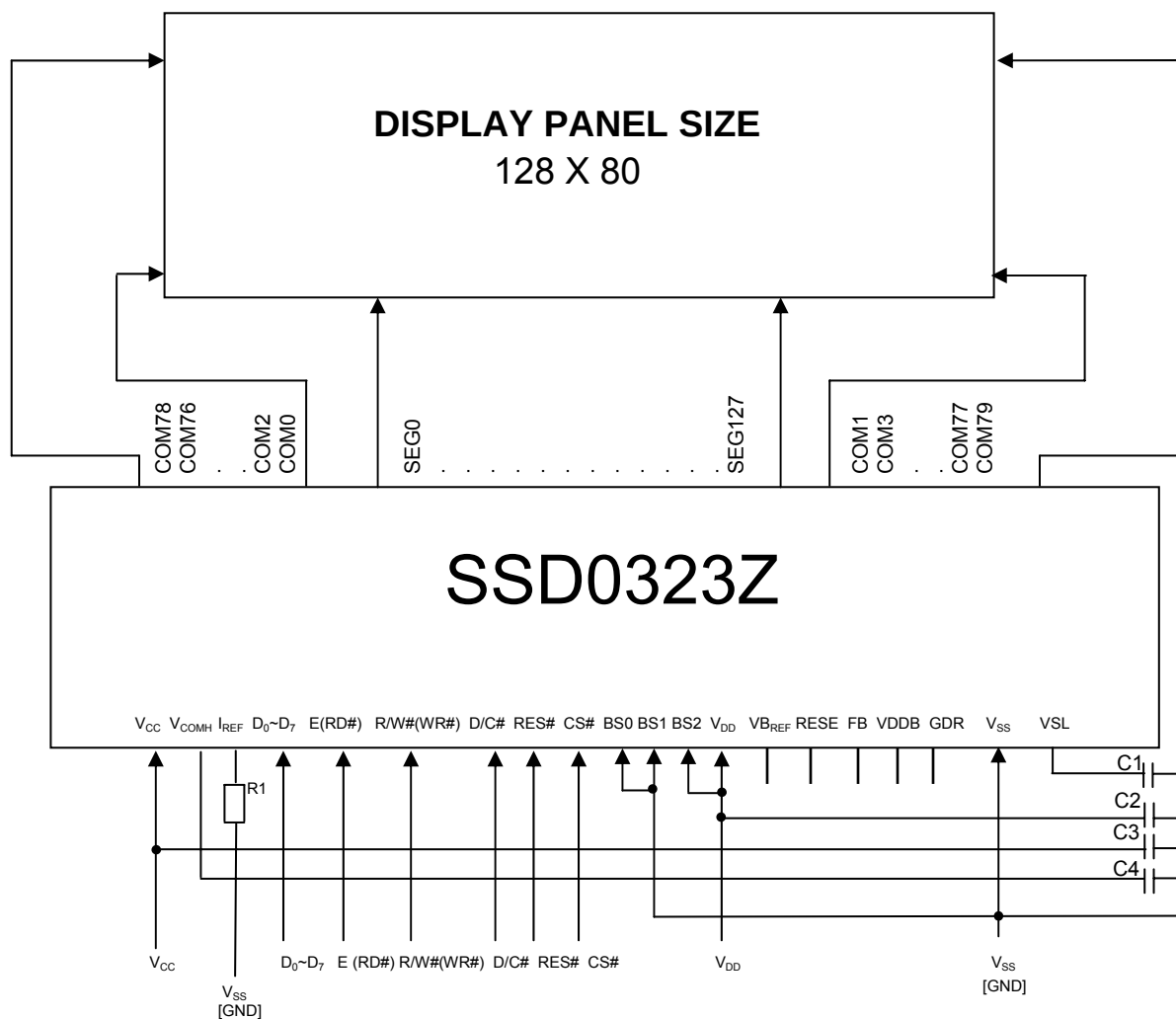


Figure 23 - Serial Interface Characteristics

APPLICATION EXAMPLE OF SSD0323Z

The configuration for 6800-parallel interface mode, externally V_{CC} is shown in the following diagram:
($V_{DD} = 3V$, $V_{CC} = V_{REF} = 12V$, $I_{REF} = 10\mu A$)



Pins connected to MCU interface: $D_0 \sim D_7$, $E(RD\#)$, $R/W\#(WR\#)$, $D/C\#$, $CS\#$, $RES\#$

Pins internally connected to V_{DD} : $M/S\#$, CLS

Pins internally connected to V_{SS} : $BS0$, $VSSB$

Pin internally connected to $V_{CC} = V_{REF}$

Pins floated: GDR , $RESE$, FB , V_{REF} , $VDDB$

$C1$: $2.2\mu F$ ⁽¹⁾

$C2 \sim C4$: $4.7\mu F$ ⁽¹⁾

$R1$: $910k\Omega$, $R1 = (Voltage\ at\ I_{REF}\ pin - V_{SS}) / I_{REF}$

Note

⁽¹⁾ The capacitor value is recommended value. Select appropriate value against module application.

Figure 24 - Application Example for SSD0323Z

SSD0323Z DIE TRAY DIMENSIONS

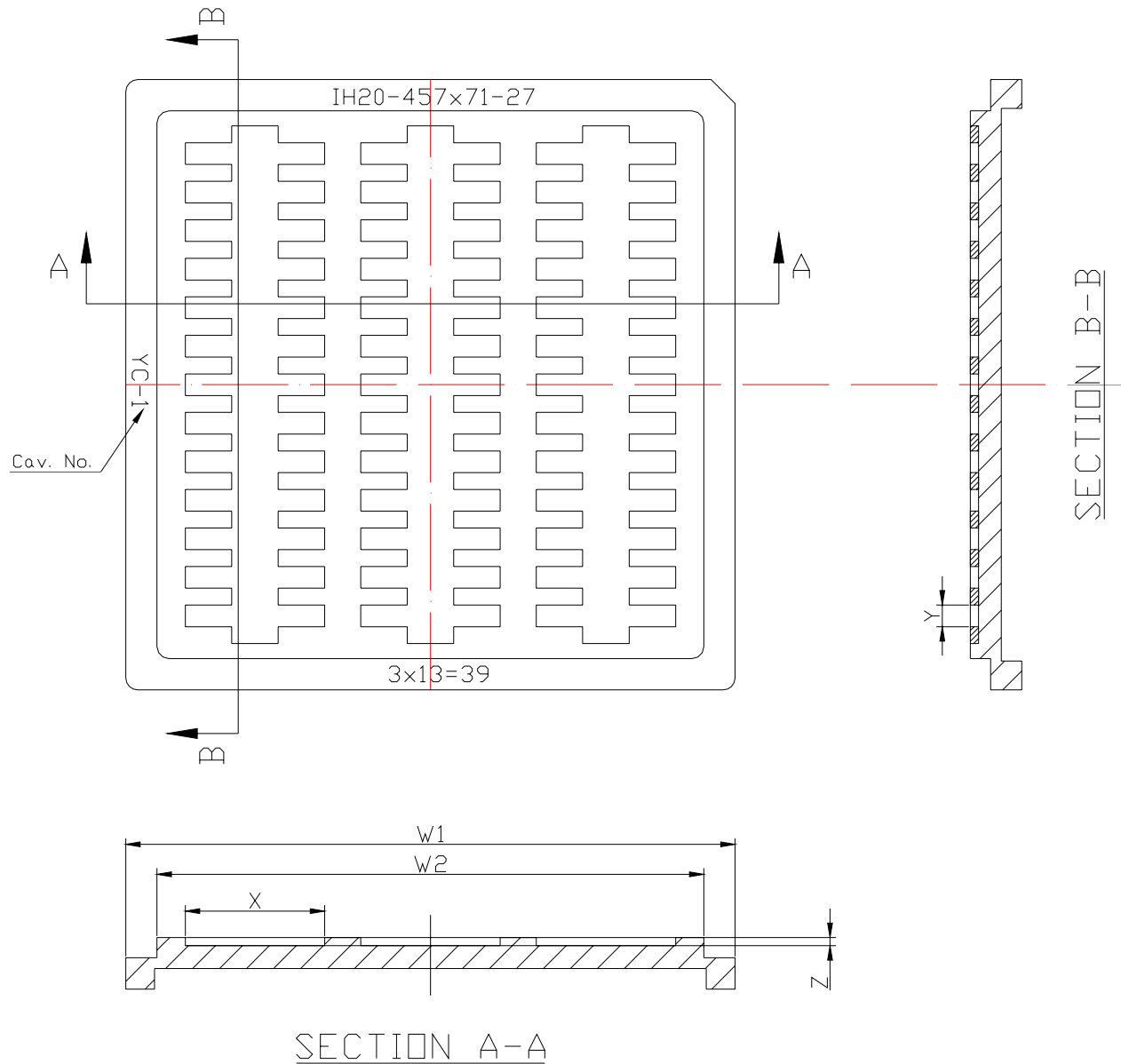


Figure 25 - SSD0323Z Die Tray Drawing

Table 28 - SSD0323Z Die Tray Dimensions

Parameter	Dimensions
W1	50.70±0.2 mm
W2	45.50±0.2 mm
X	11.60±0.1 mm
Y	1.80±0.1 mm
Z	0.71±0.05 mm
N (number of die)	39

SSD1325T6R1 PACKAGE DETAILS

Note
(1) The SSD1325T6R1 package details refer to SSD1325 rev1.6 datasheet.

SSD1325T6R1 Pin Assignment

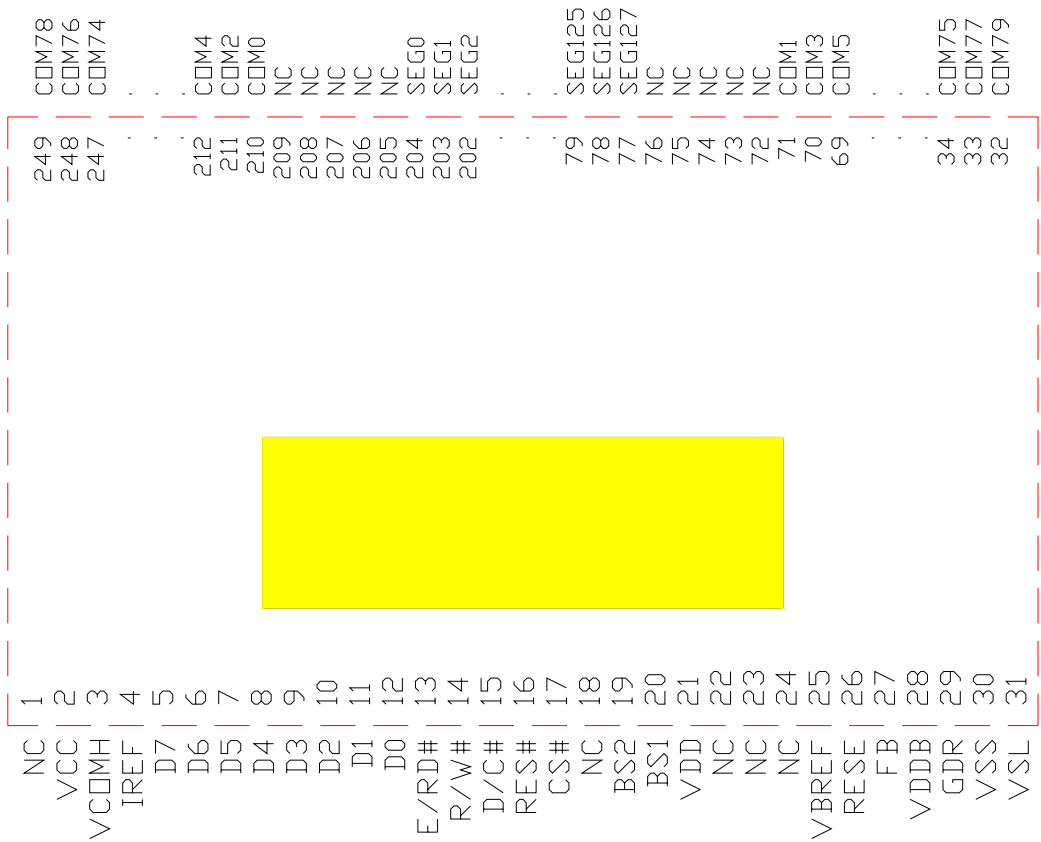


Table 29 - SSD1325T6R1 pin assignment

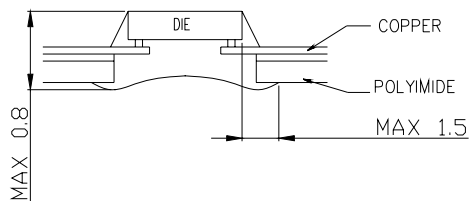
PIN NO.	PIN NAME	PIN NO.	PIN NAME	PIN NO.	PIN NAME	PIN NO.	PIN NAME
1	NC	81	SEG123	161	SEG43	241	COM62
2	VCC	82	SEG122	162	SEG42	242	COM64
3	VCOMH	83	SEG121	163	SEG41	243	COM66
4	IREF	84	SEG120	164	SEG40	244	COM68
5	D7	85	SEG119	165	SEG39	245	COM70
6	D6	86	SEG118	166	SEG38	246	COM72
7	D5	87	SEG117	167	SEG37	247	COM74
8	D4	88	SEG116	168	SEG36	248	COM76
9	D3	89	SEG115	169	SEG35	249	COM78
10	D2	90	SEG114	170	SEG34		
11	D1	91	SEG113	171	SEG33		
12	D0	92	SEG112	172	SEG32		
13	E/RD#	93	SEG111	173	SEG31		
14	R/W#	94	SEG110	174	SEG30		
15	D/C#	95	SEG109	175	SEG29		
16	RES#	96	SEG108	176	SEG28		
17	CS#	97	SEG107	177	SEG27		
18	NC	98	SEG106	178	SEG26		
19	BS2	99	SEG105	179	SEG25		
20	BS1	100	SEG104	180	SEG24		
21	VDD	101	SEG103	181	SEG23		
22	NC	102	SEG102	182	SEG22		
23	NC	103	SEG101	183	SEG21		
24	NC	104	SEG100	184	SEG20		
25	VBREF	105	SEG99	185	SEG19		
26	RESE	106	SEG98	186	SEG18		
27	FR	107	SEG97	187	SEG17		
28	VDDb	108	SEG96	188	SEG16		
29	GDR	109	SEG95	189	SEG15		
30	VSS	110	SEG94	190	SEG14		
31	VSL	111	SEG93	191	SEG13		
32	COM9	112	SEG92	192	SEG12		
33	COM7	113	SEG91	193	SEG11		
34	COM5	114	SEG90	194	SEG10		
35	COM3	115	SEG89	195	SEG9		
36	COM1	116	SEG88	196	SEG8		
37	COM9	117	SEG87	197	SEG7		
38	COM7	118	SEG86	198	SEG6		
39	COM5	119	SEG85	199	SEG5		
40	COM3	120	SEG84	200	SEG4		
41	COM1	121	SEG83	201	SEG3		
42	COM9	122	SEG82	202	SEG2		
43	COM7	123	SEG81	203	SEG1		
44	COM5	124	SEG80	204	SEG0		
45	COM3	125	SEG79	205	NC		
46	COM1	126	SEG78	206	NC		
47	COM9	127	SEG77	207	NC		
48	COM7	128	SEG76	208	NC		
49	COM5	129	SEG75	209	NC		
50	COM3	130	SEG74	210	COM0		
51	COM1	131	SEG73	211	COM2		
52	COM9	132	SEG72	212	COM4		
53	COM7	133	SEG71	213	COM6		
54	COM5	134	SEG70	214	COM8		
55	COM3	135	SEG69	215	COM10		
56	COM1	136	SEG68	216	COM12		
57	COM9	137	SEG67	217	COM14		
58	COM7	138	SEG66	218	COM16		
59	COM5	139	SEG65	219	COM18		
60	COM3	140	SEG64	220	COM20		
61	COM1	141	SEG63	221	COM22		
62	COM9	142	SEG62	222	COM24		
63	COM7	143	SEG61	223	COM26		
64	COM5	144	SEG60	224	COM28		
65	COM3	145	SEG59	225	COM30		
66	COM1	146	SEG58	226	COM32		
67	COM9	147	SEG57	227	COM34		
68	COM7	148	SEG56	228	COM36		
69	COM5	149	SEG55	229	COM38		
70	COM3	150	SEG54	230	COM40		
71	COM1	151	SEG53	231	COM42		
72	NC	152	SEG52	232	COM44		
73	NC	153	SEG51	233	COM46		
74	NC	154	SEG50	234	COM48		
75	NC	155	SEG49	235	COM50		
76	NC	156	SEG48	236	COM52		
77	SEG127	157	SEG47	237	COM54		
78	SEG126	158	SEG46	238	COM56		
79	SEG125	159	SEG45	239	COM58		
80	SEG124	160	SEG44	240	COM60		

SSD0323

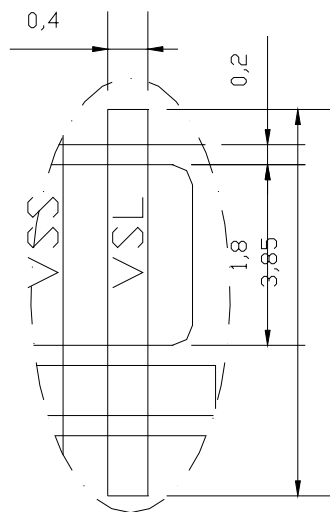
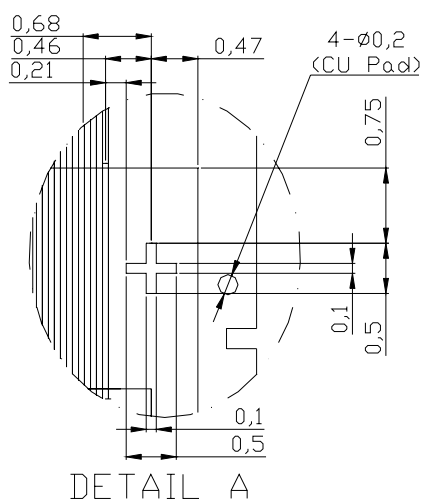


NOTE:

1. GENERAL TOLERANCE: $\pm 0.05\text{mm}$
2. CUTLINE TOLERANCE: $\pm 0.15\text{mm}$
3. MATERIAL
 - PI: $75\pm 6\mu\text{m}$
 - CU: $18\pm 5\mu\text{m}$
 - SR: $26\pm 14\mu\text{m}$
 - ADHESIVE: $12\pm 2\mu\text{m}$
4. FLEX COATING: $\text{Min} 10\mu\text{m}$
5. SN PLATING: $0.20\pm 0.05\mu\text{m}$
6. TAP SITE: 4 SPH, 19.00mm



MIRROR DESIGN



DETAIL B

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